

Exact POMDP Solutions: α -vectors

Recap

Recap

- POMDP

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

$$b_t(s) = P(s_t = s \mid h_t)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

$$b_t(s) = P(s_t = s \mid h_t)$$

$$b' = \tau(b, a, o)$$

Recap

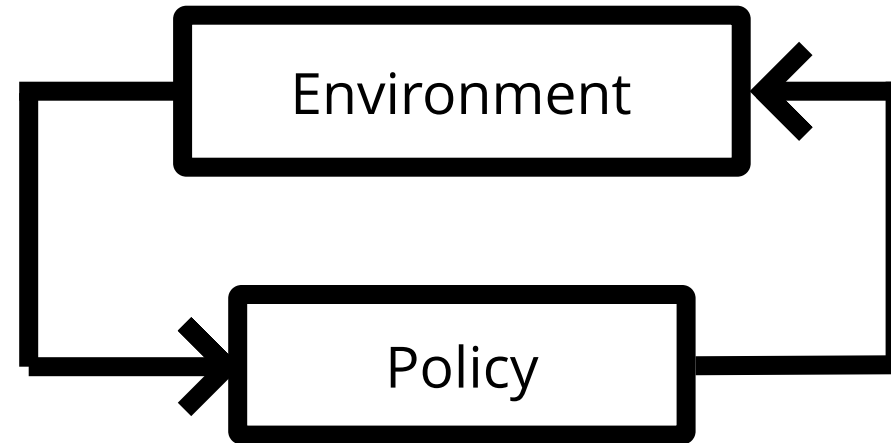
- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

$$b_t(s) = P(s_t = s \mid h_t)$$

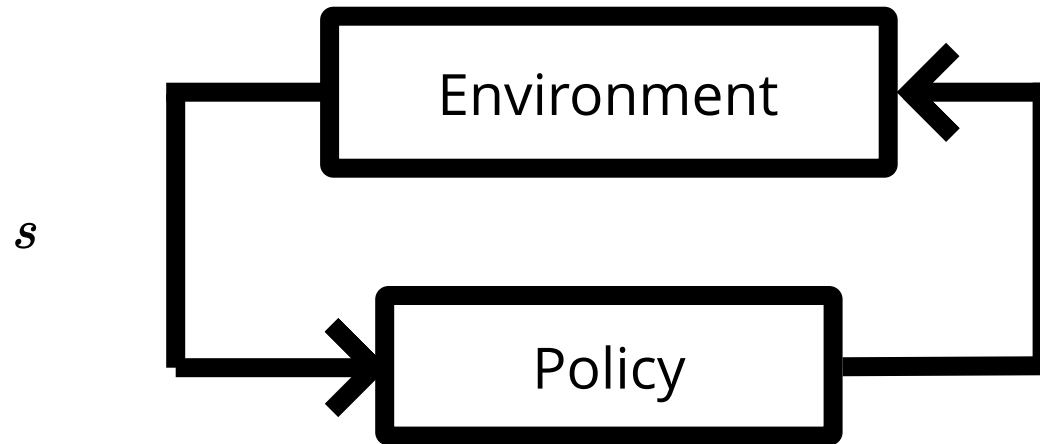
$$b' = \tau(b, a, o)$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

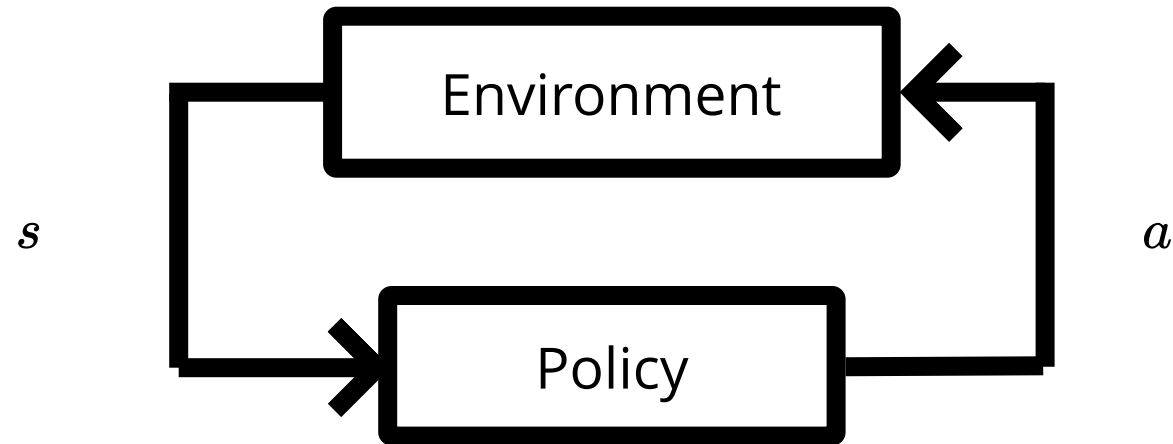
MDP Sense-Plan-Act Loop



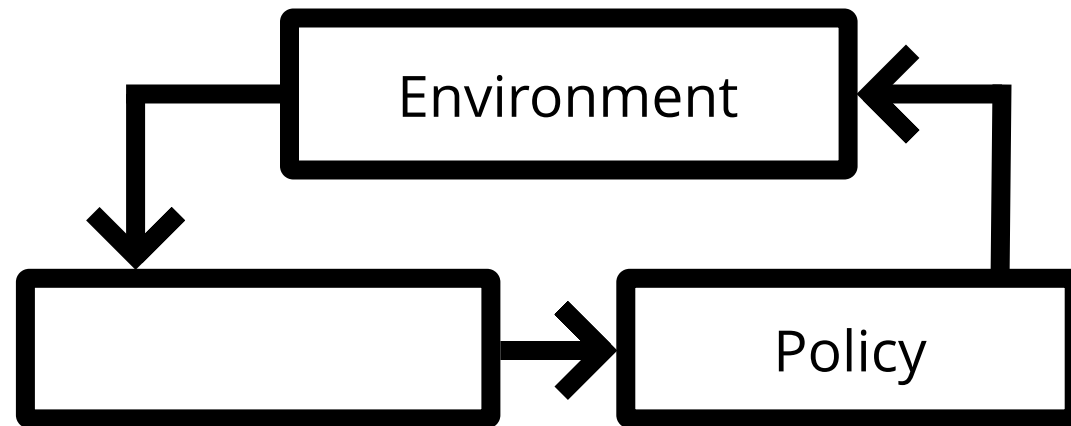
MDP Sense-Plan-Act Loop



MDP Sense-Plan-Act Loop



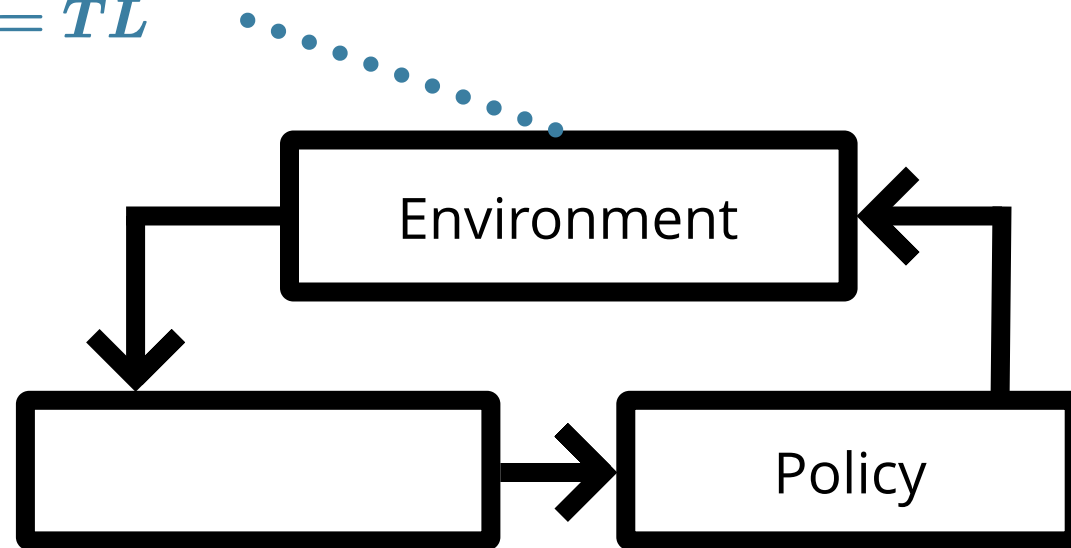
POMDP Sense-Plan-Act Loop



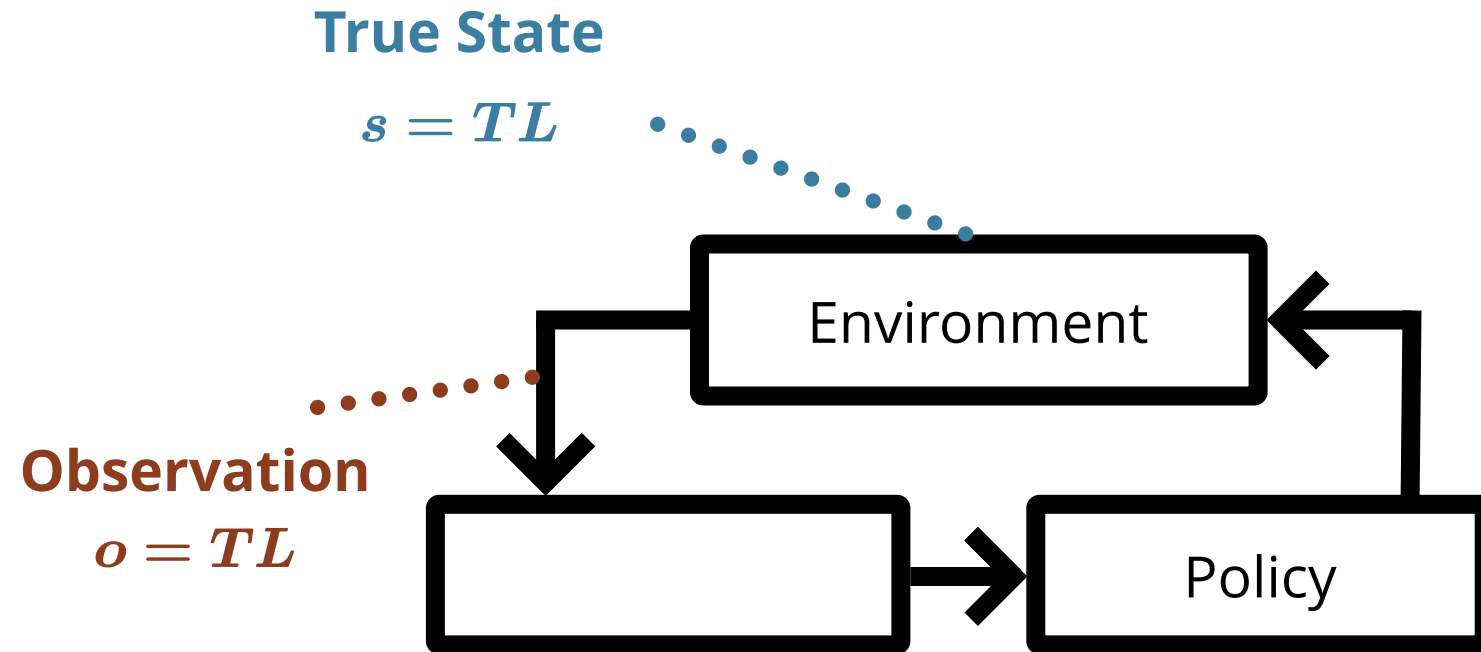
POMDP Sense-Plan-Act Loop

True State

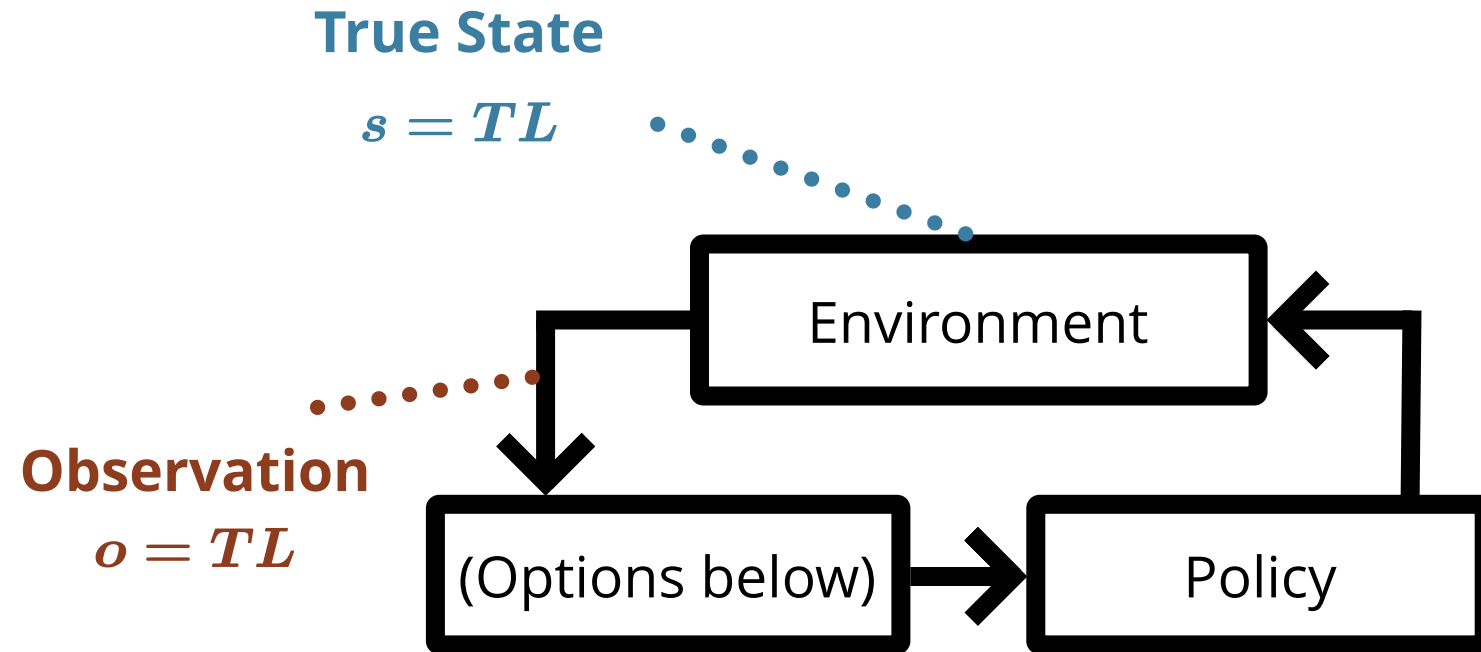
$$s = TL$$



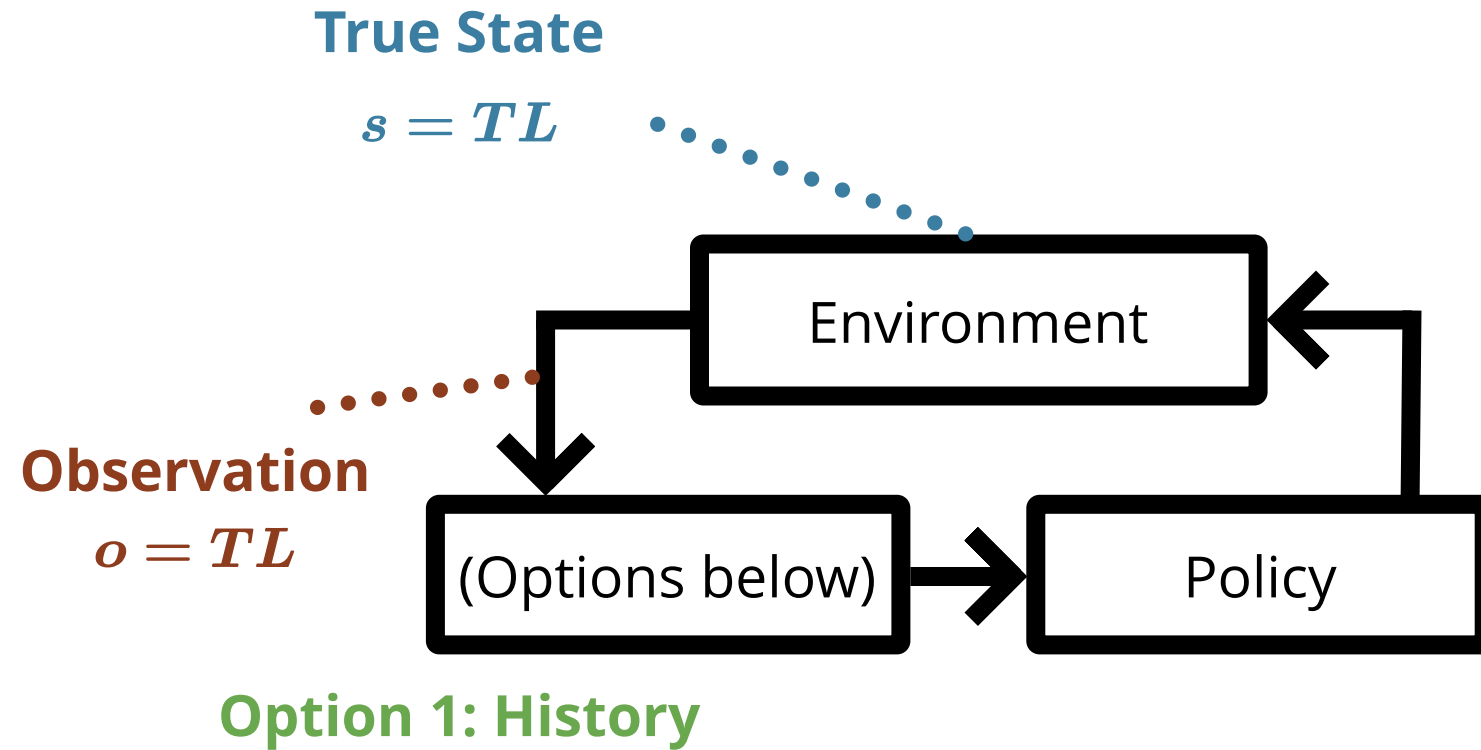
POMDP Sense-Plan-Act Loop



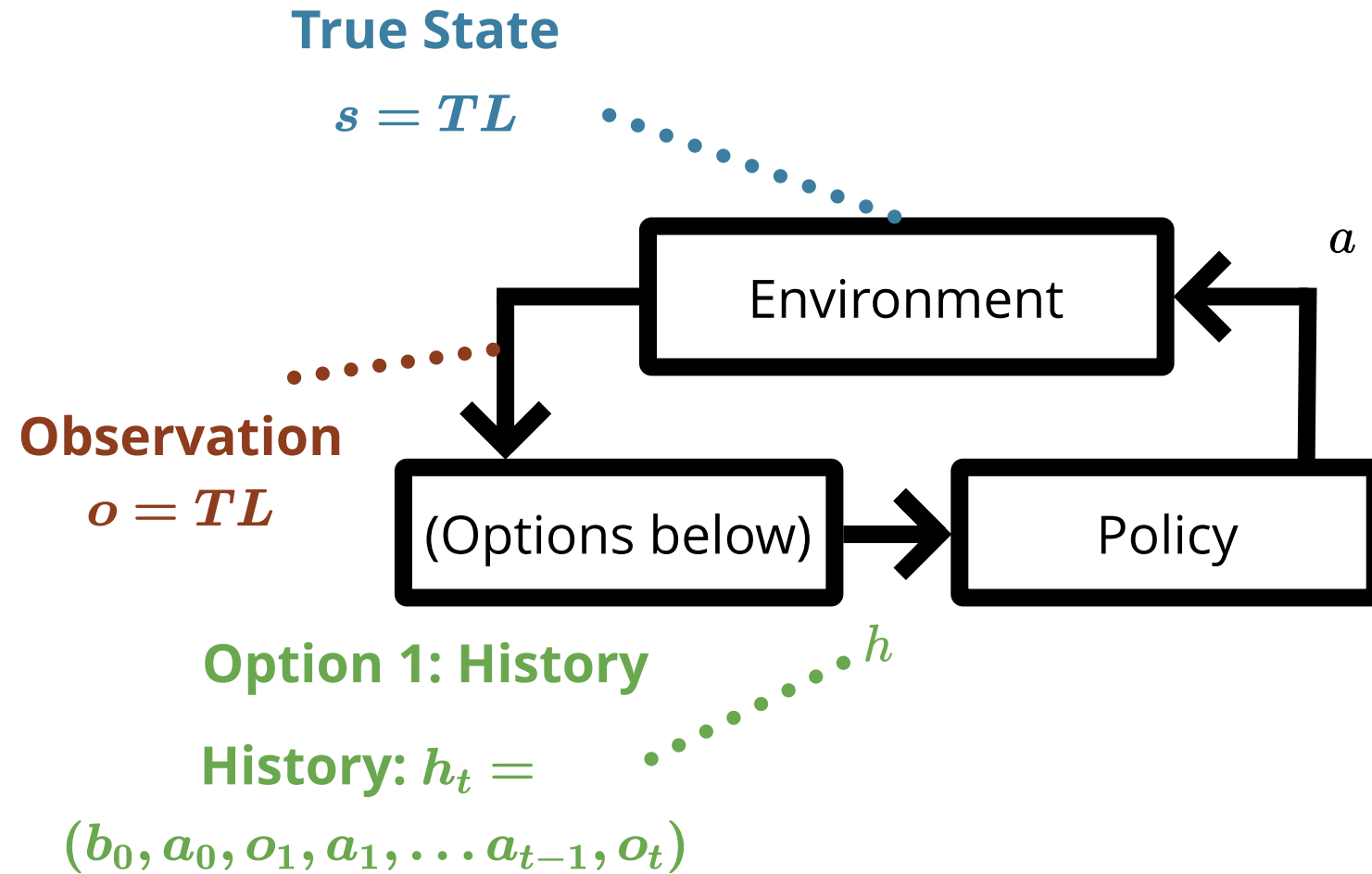
POMDP Sense-Plan-Act Loop



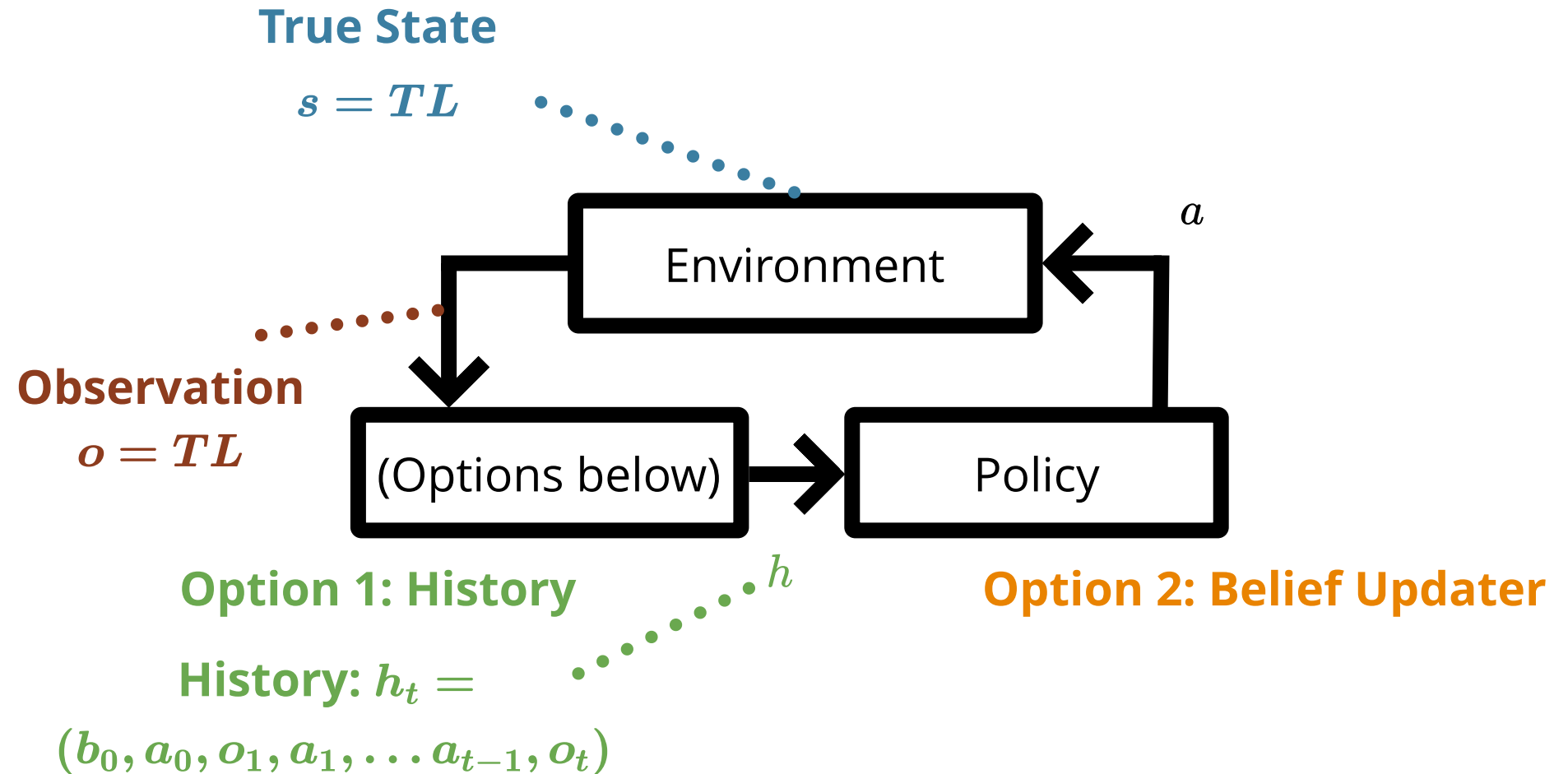
POMDP Sense-Plan-Act Loop



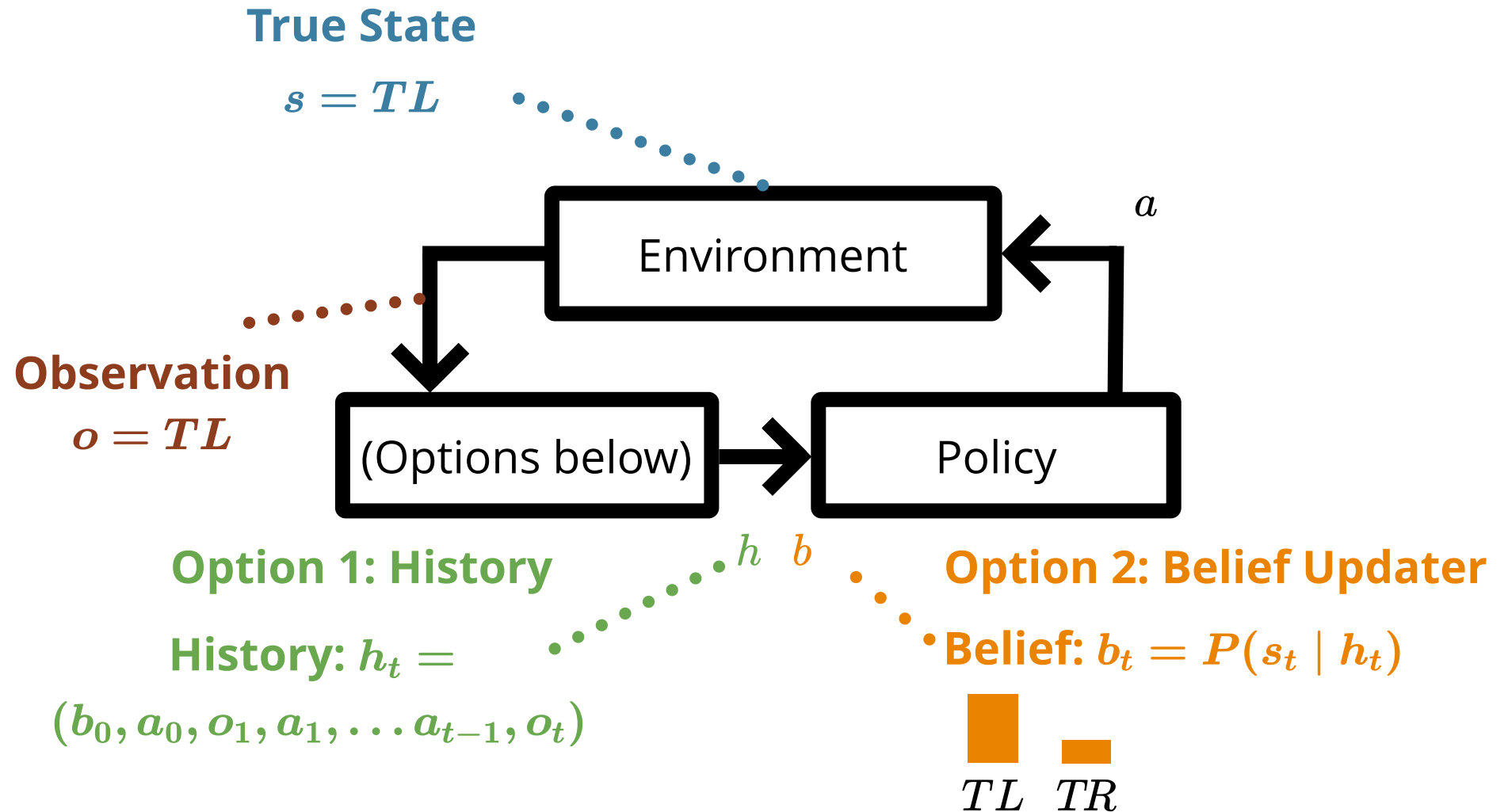
POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



Exercise 1: Crying Baby Belief Update

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

$$R(s, a) = R(s) + R(a)$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

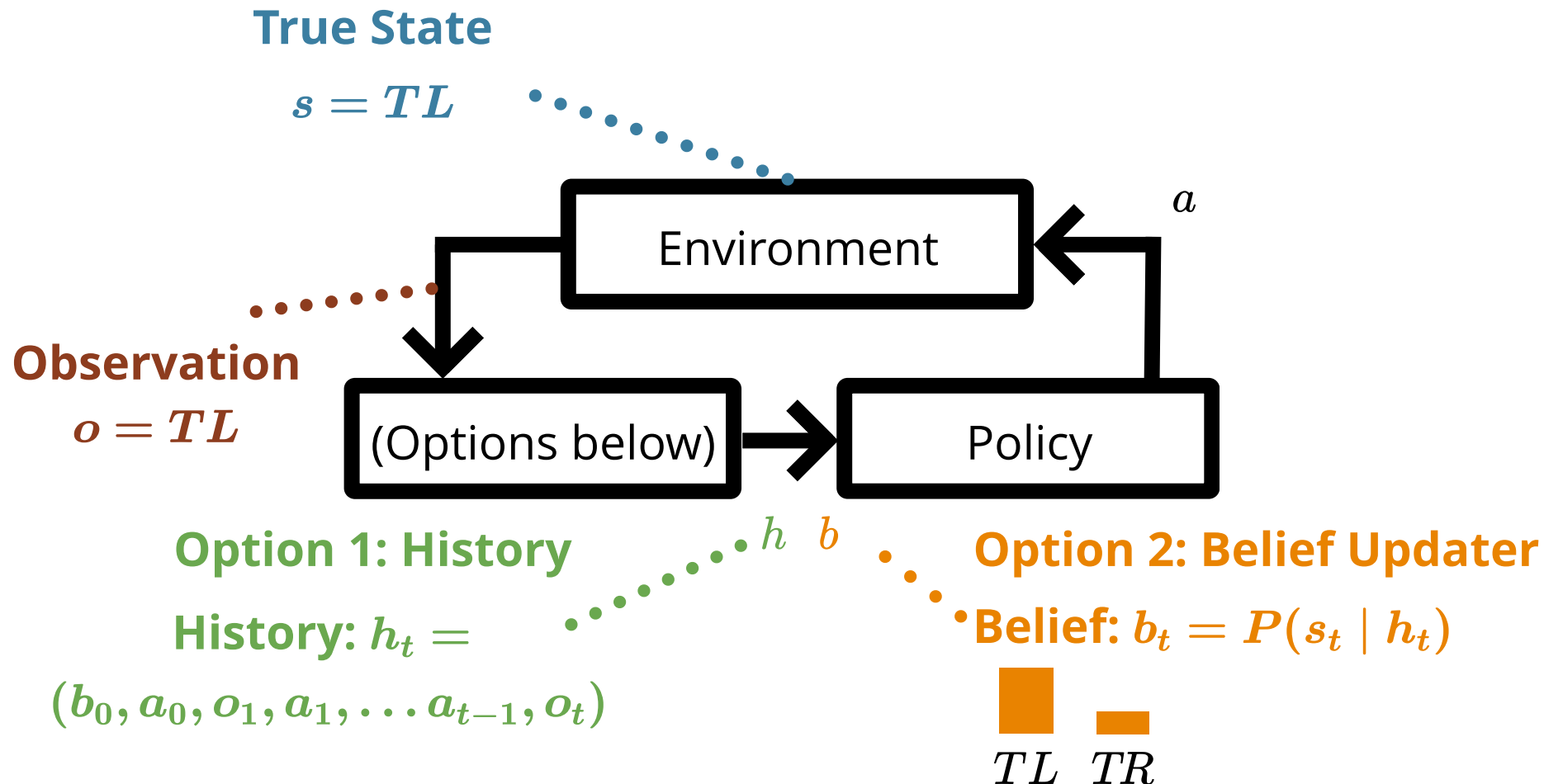
$$\gamma = 0.9$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

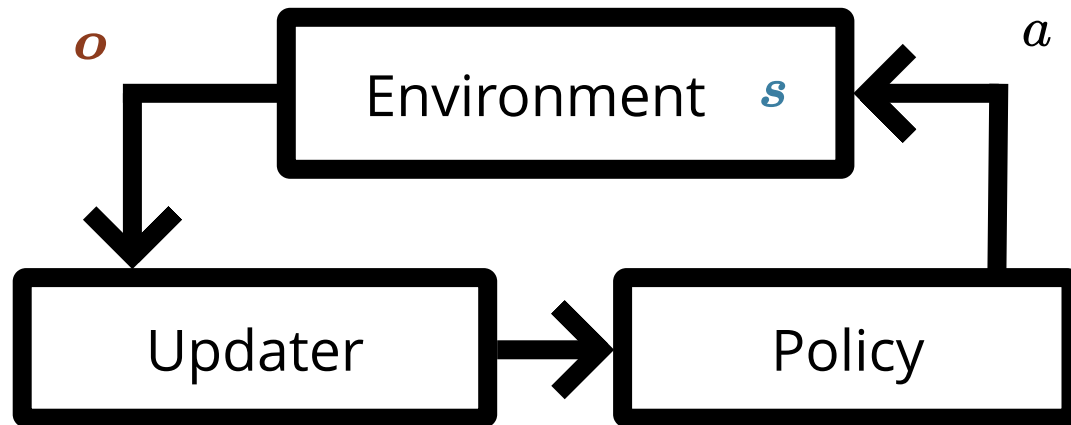
Starting at a $b(h) = 0$, calculate b' with $a = \neg f$ and $o = c$.

Guiding Quesiton

How do we calculate the optimal action in a POMDP?



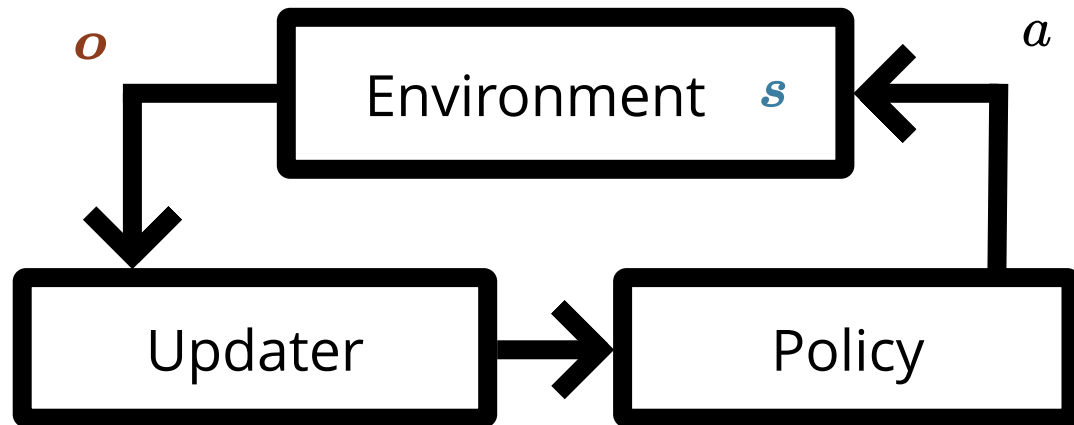
Belief-Space MDP



$$b_t = P(s_t | h_t)$$

 TL  TR

Belief-Space MDP

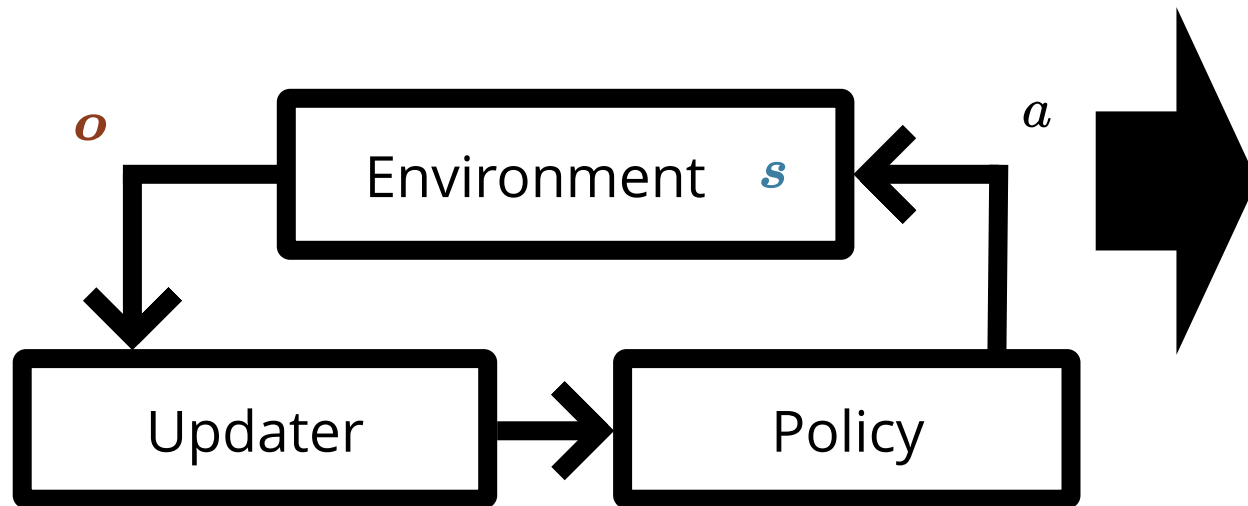


$$b_t = P(s_t | h_t)$$


TL


TR

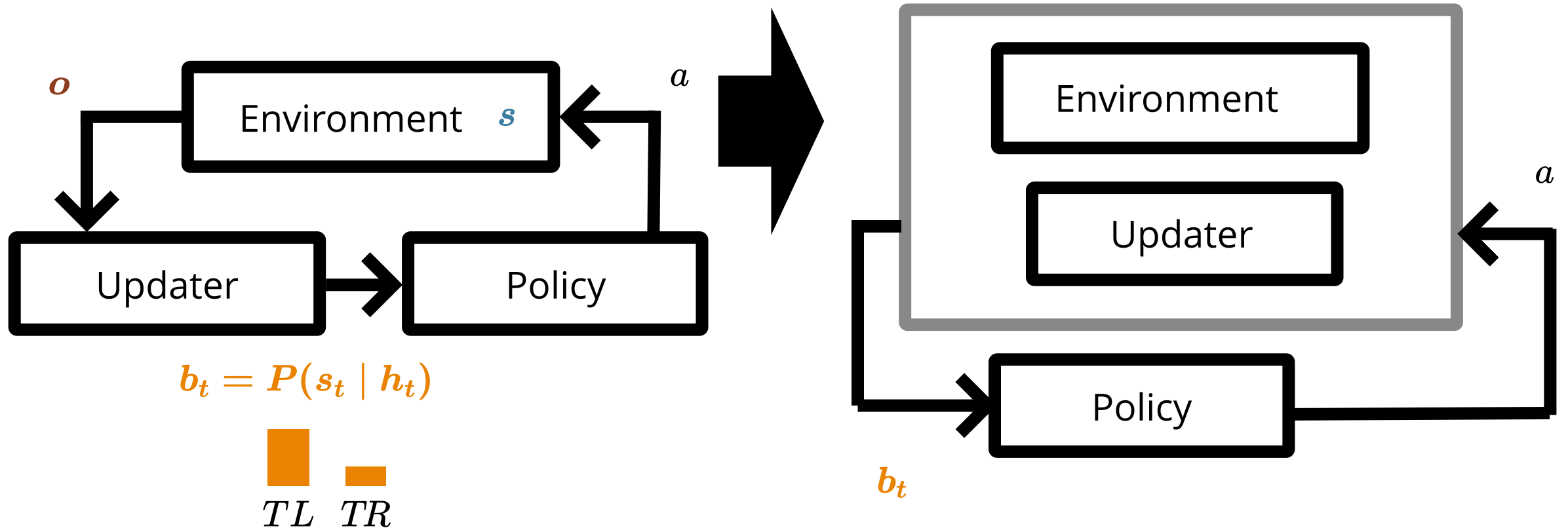
Belief-Space MDP



$$b_t = P(s_t | h_t)$$

TL TR

Belief-Space MDP



Belief Dynamics

Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

Belief Dynamics

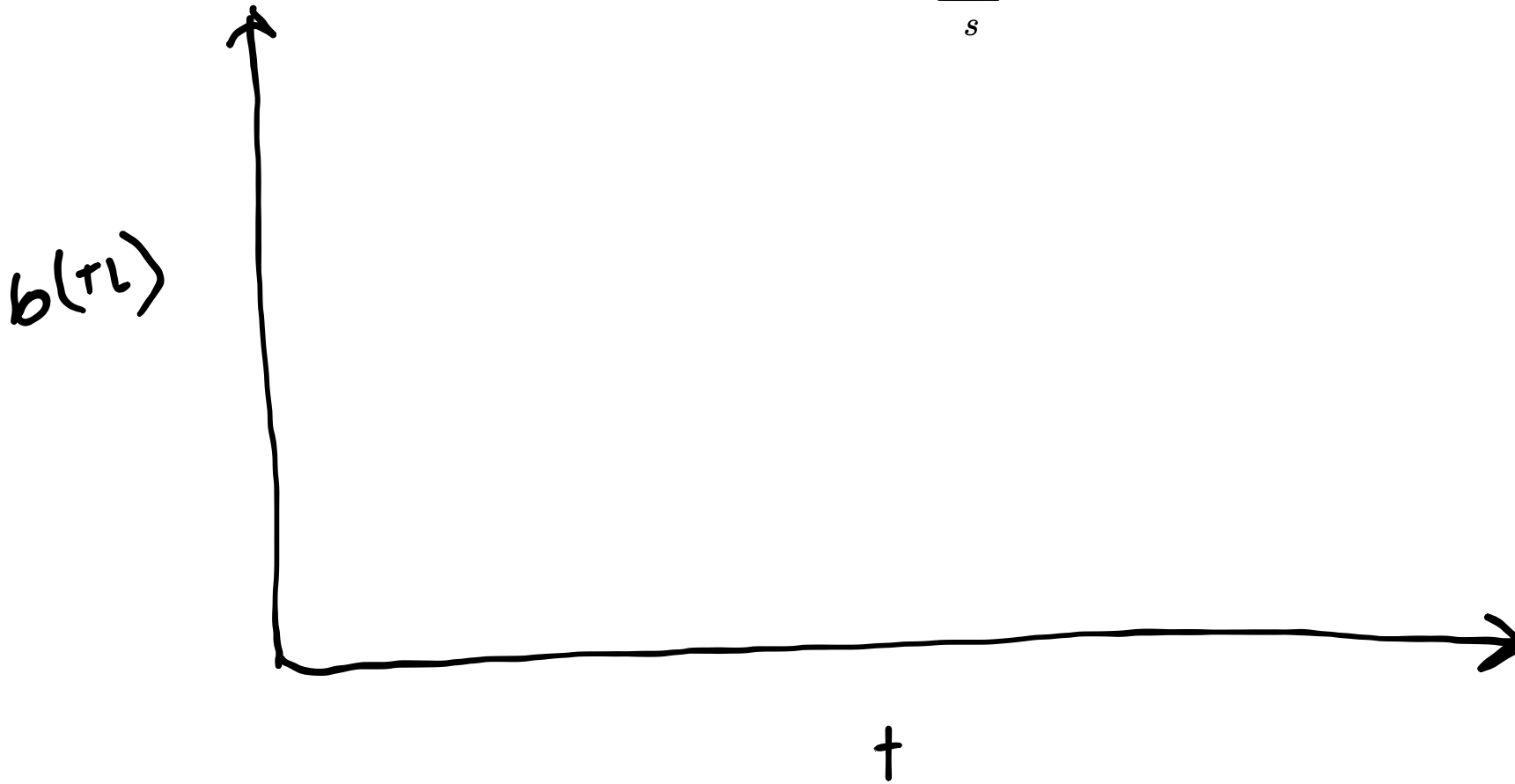
$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$b(\tau_L)$

†

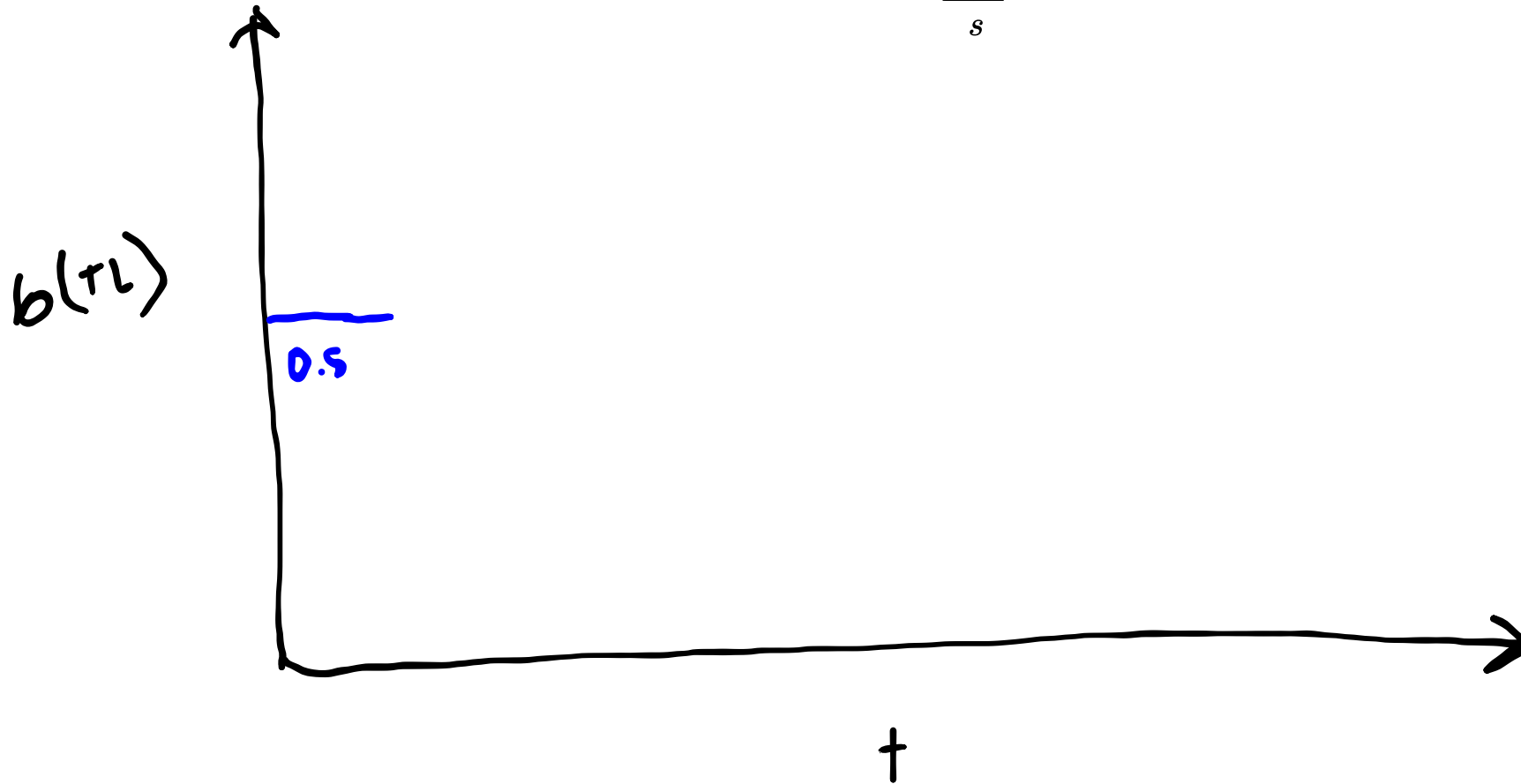
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



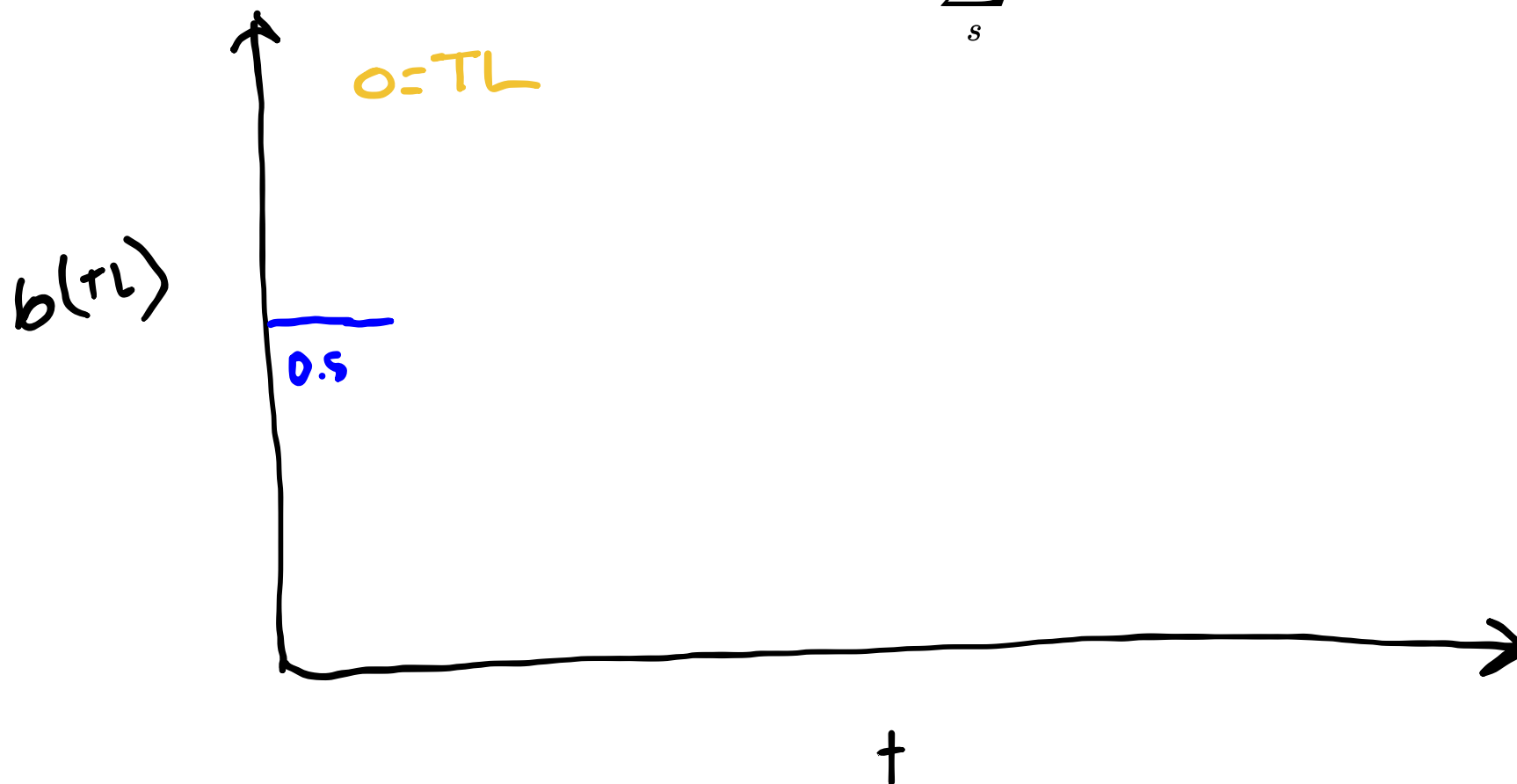
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



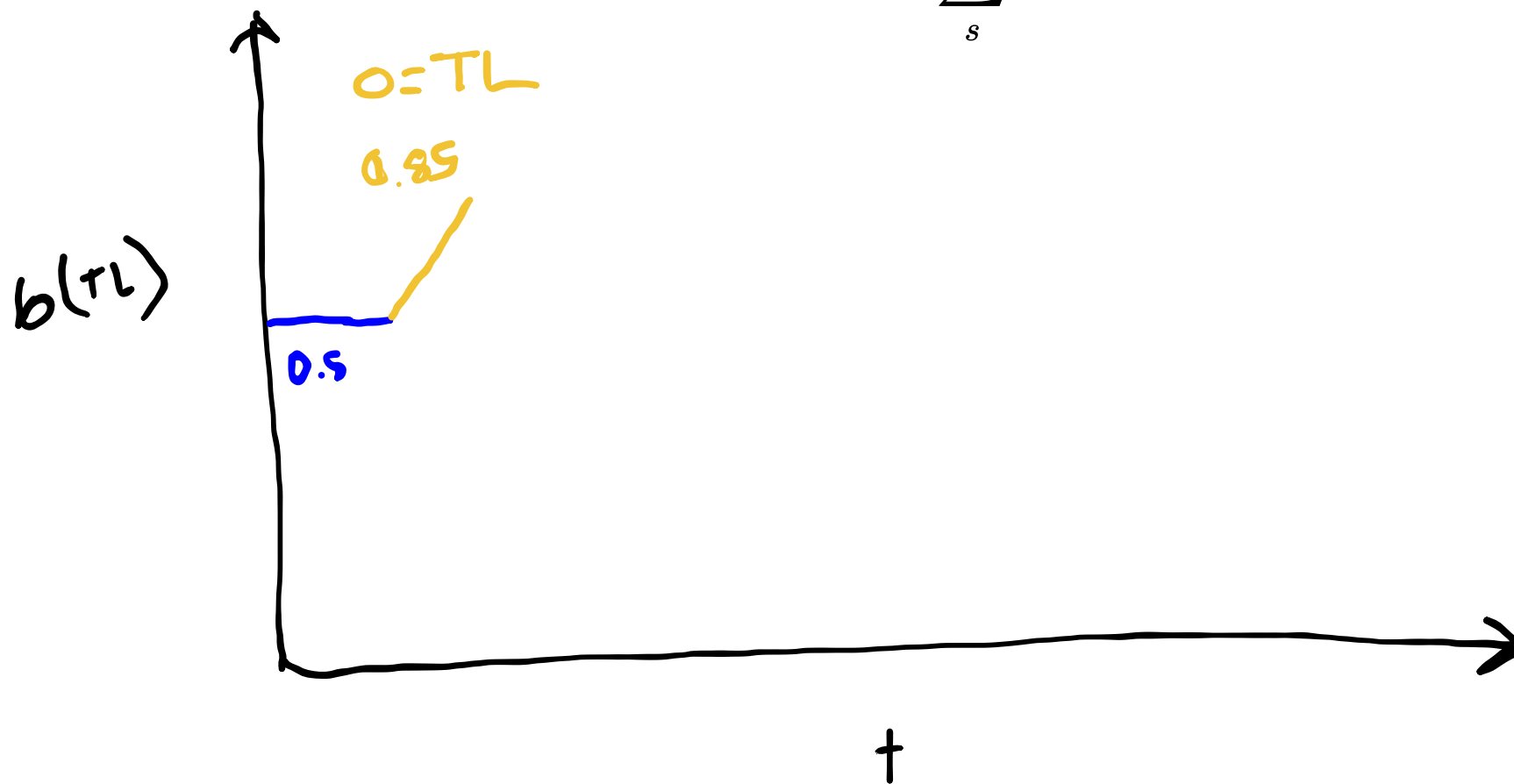
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



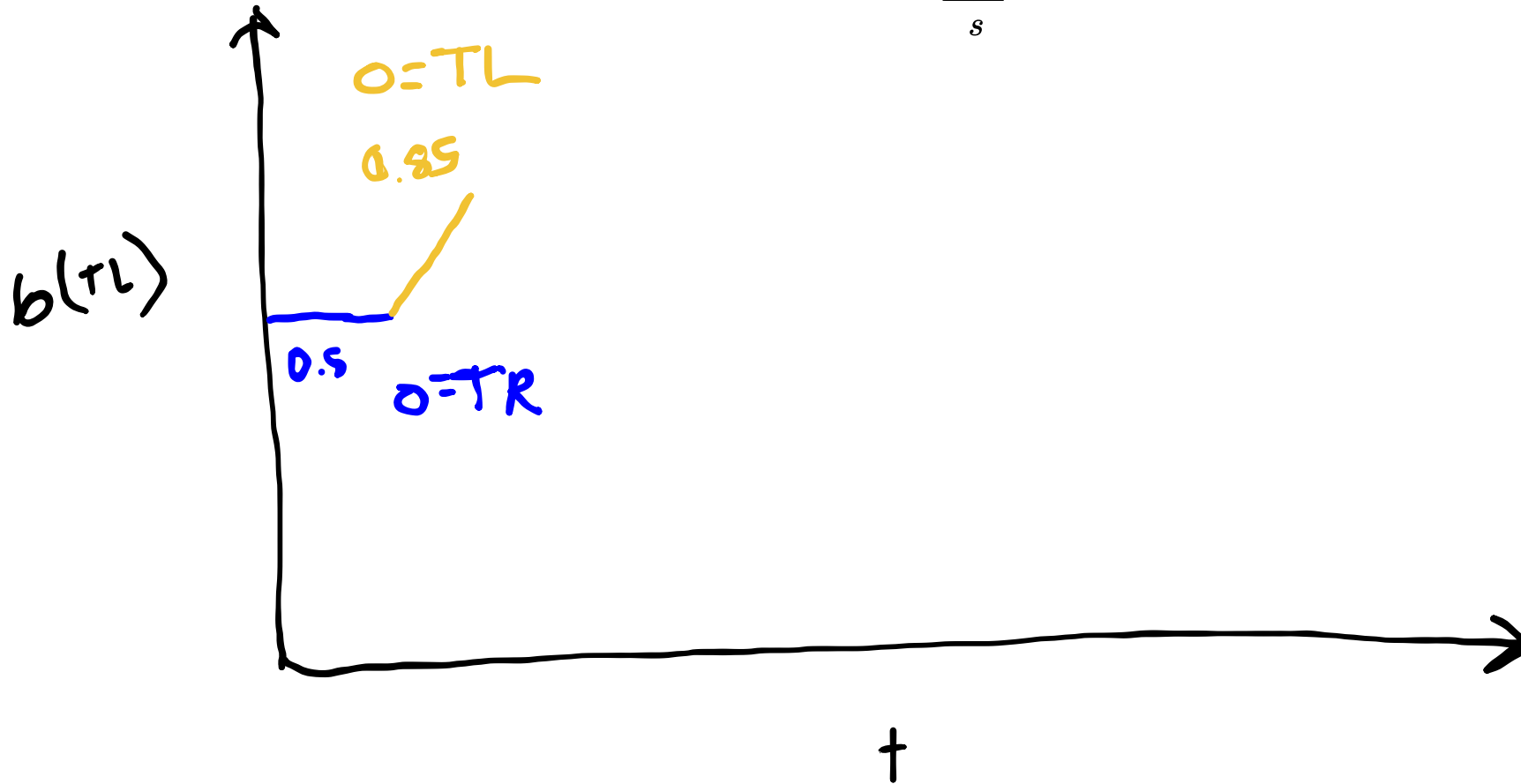
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



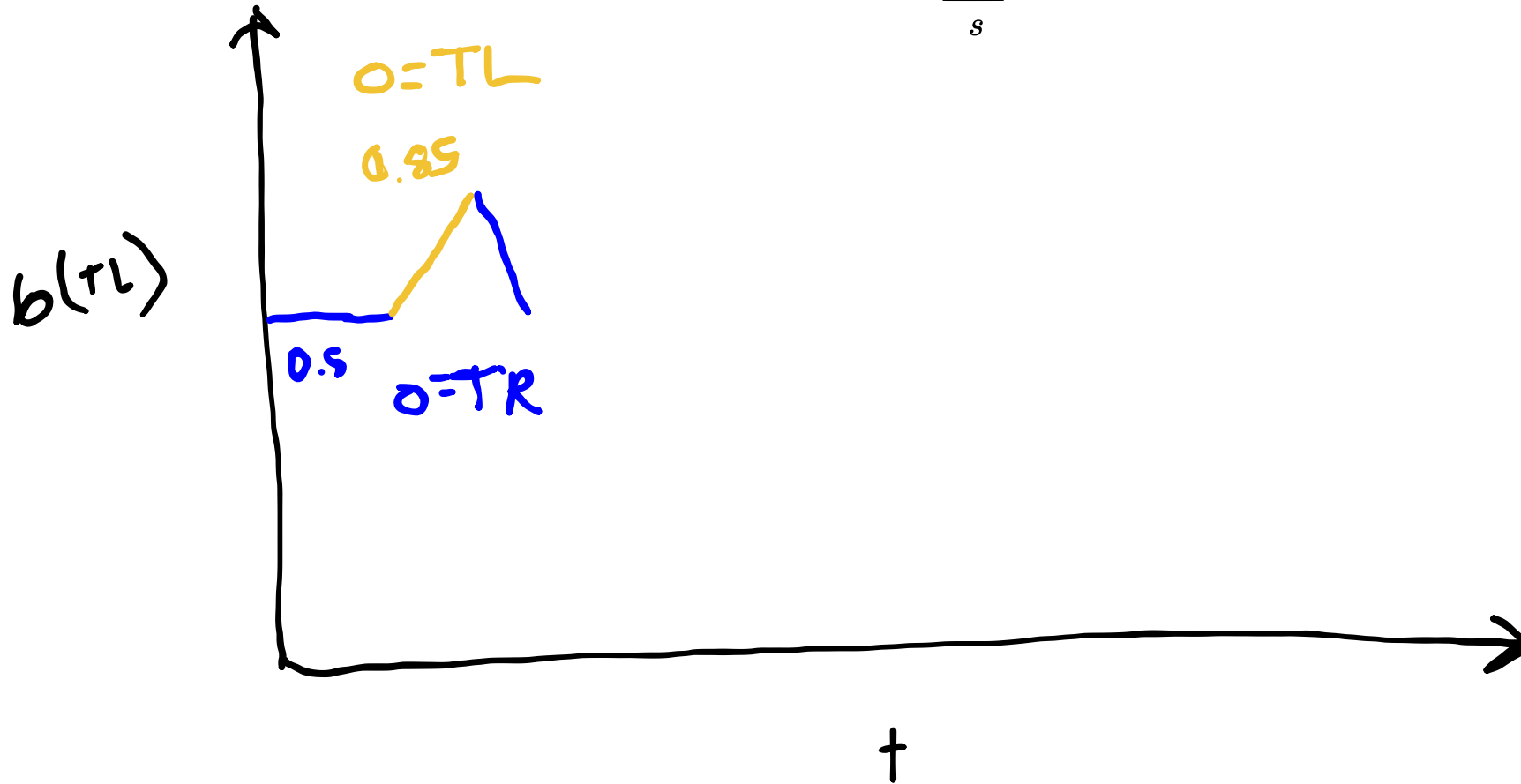
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



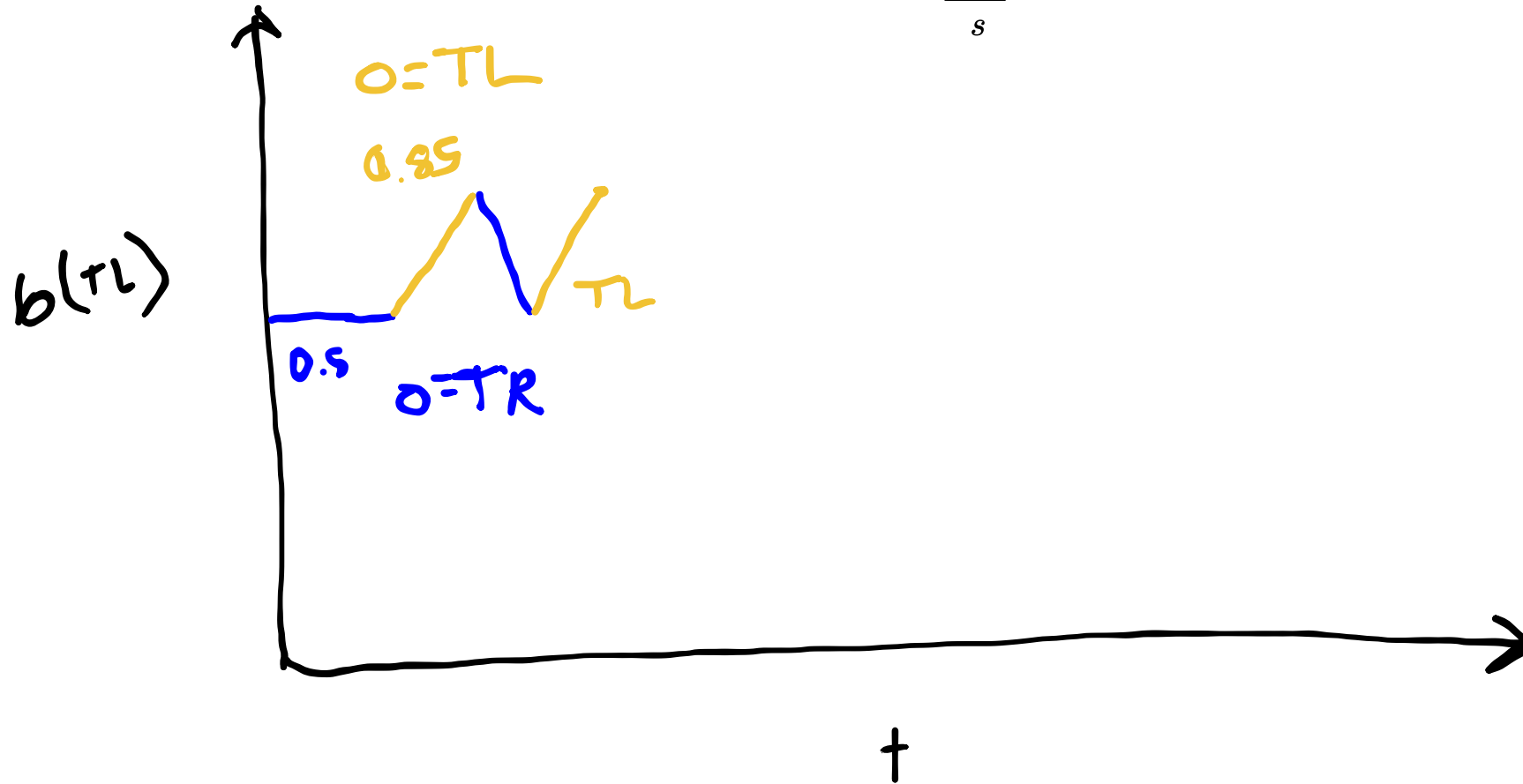
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



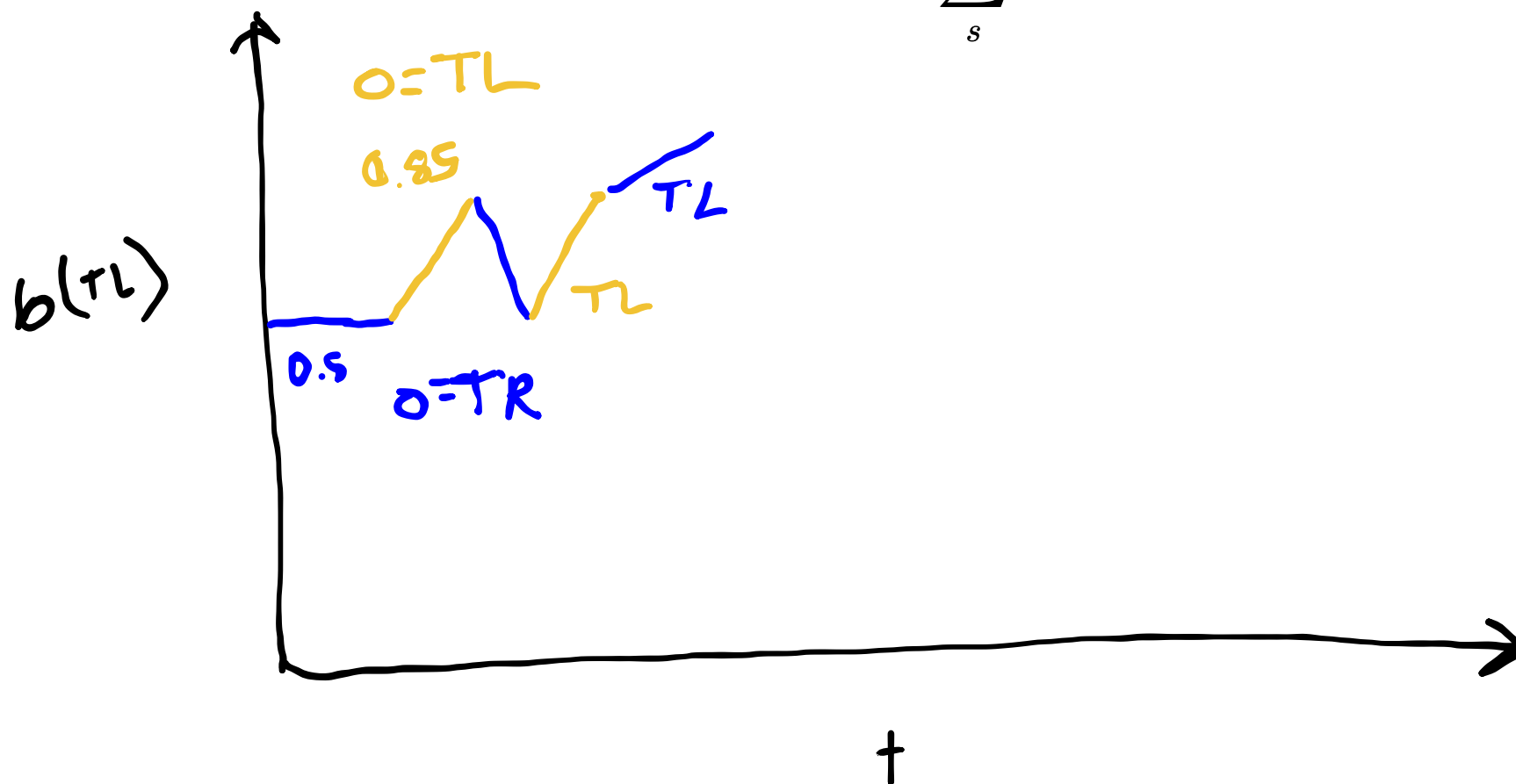
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



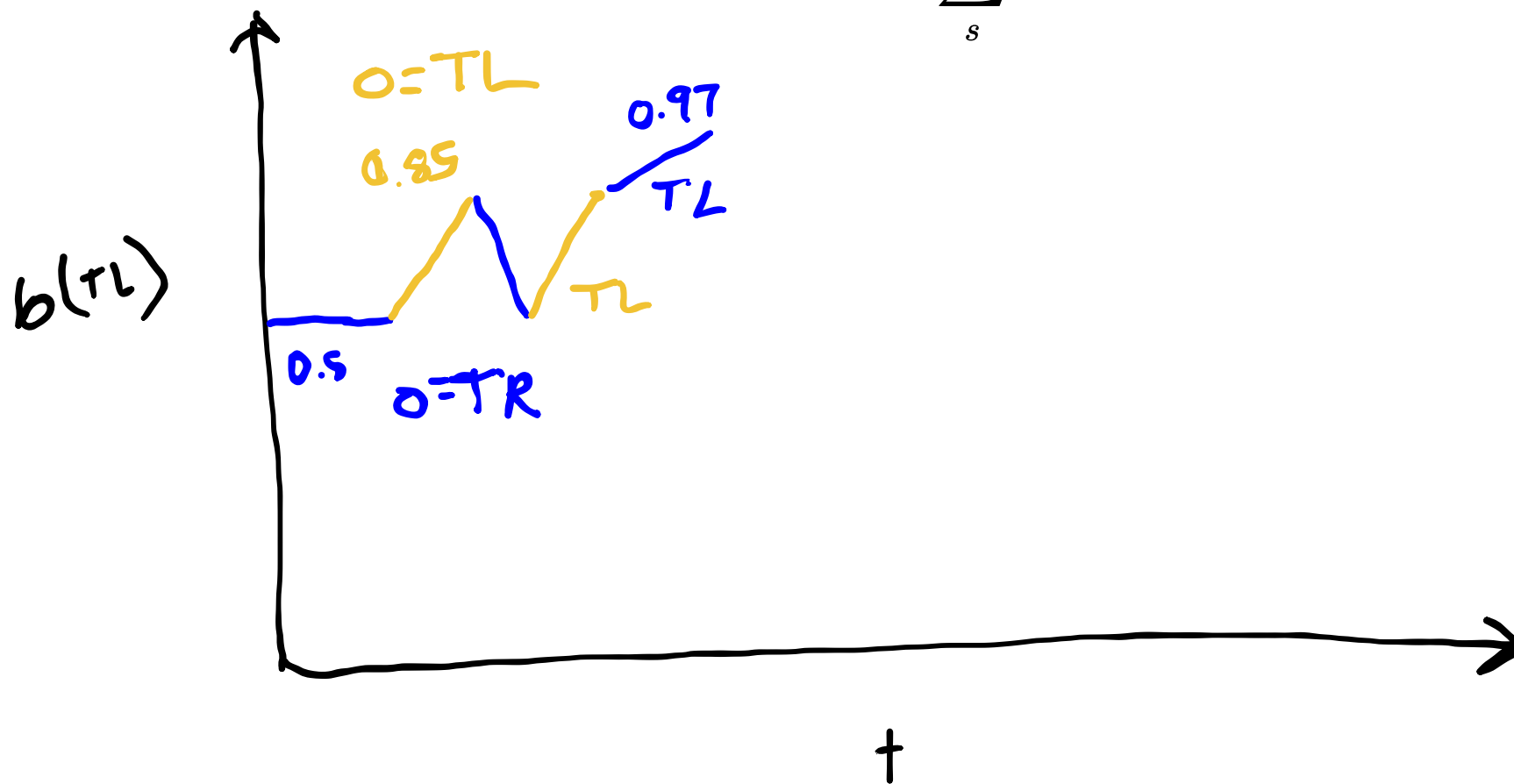
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



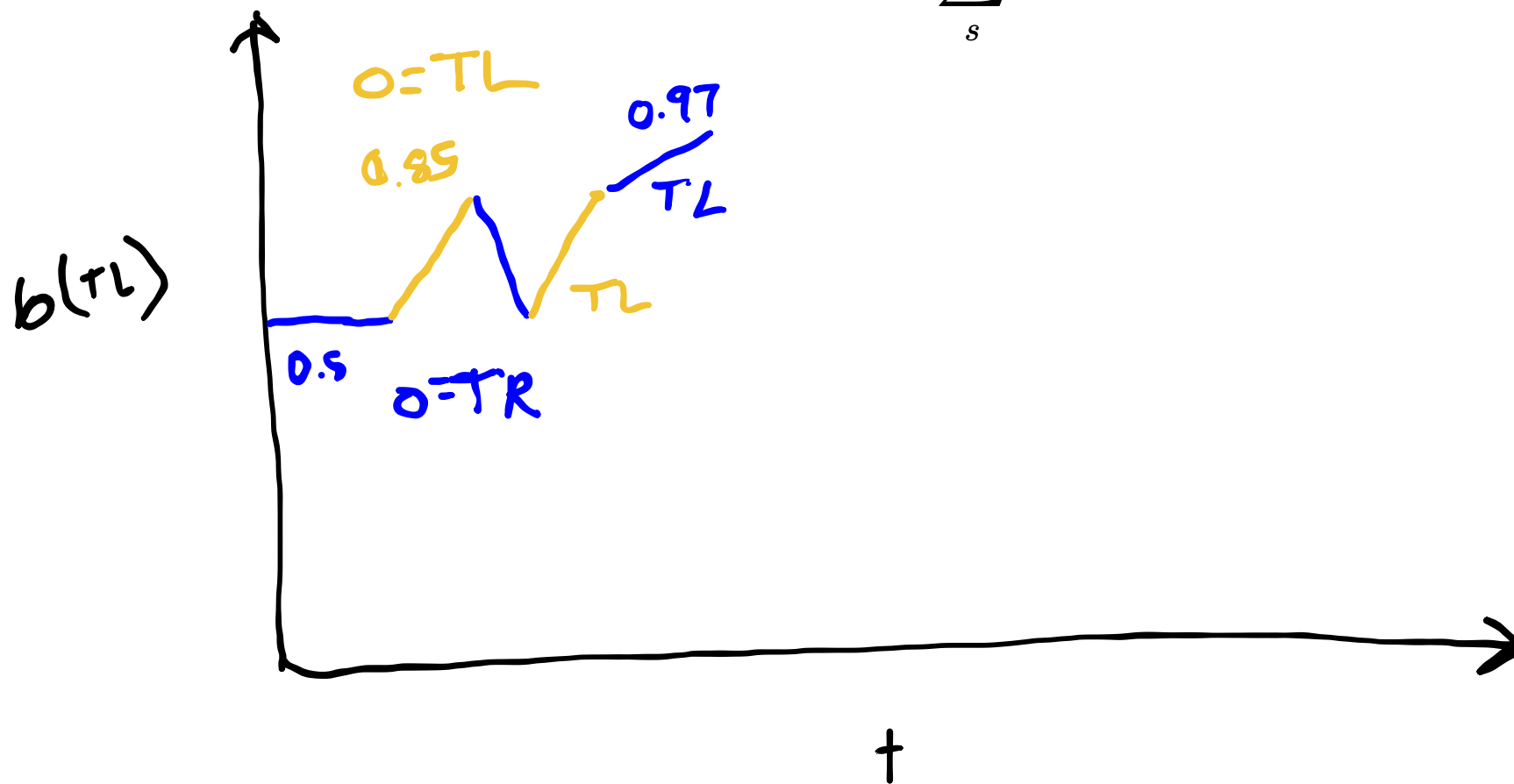
Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



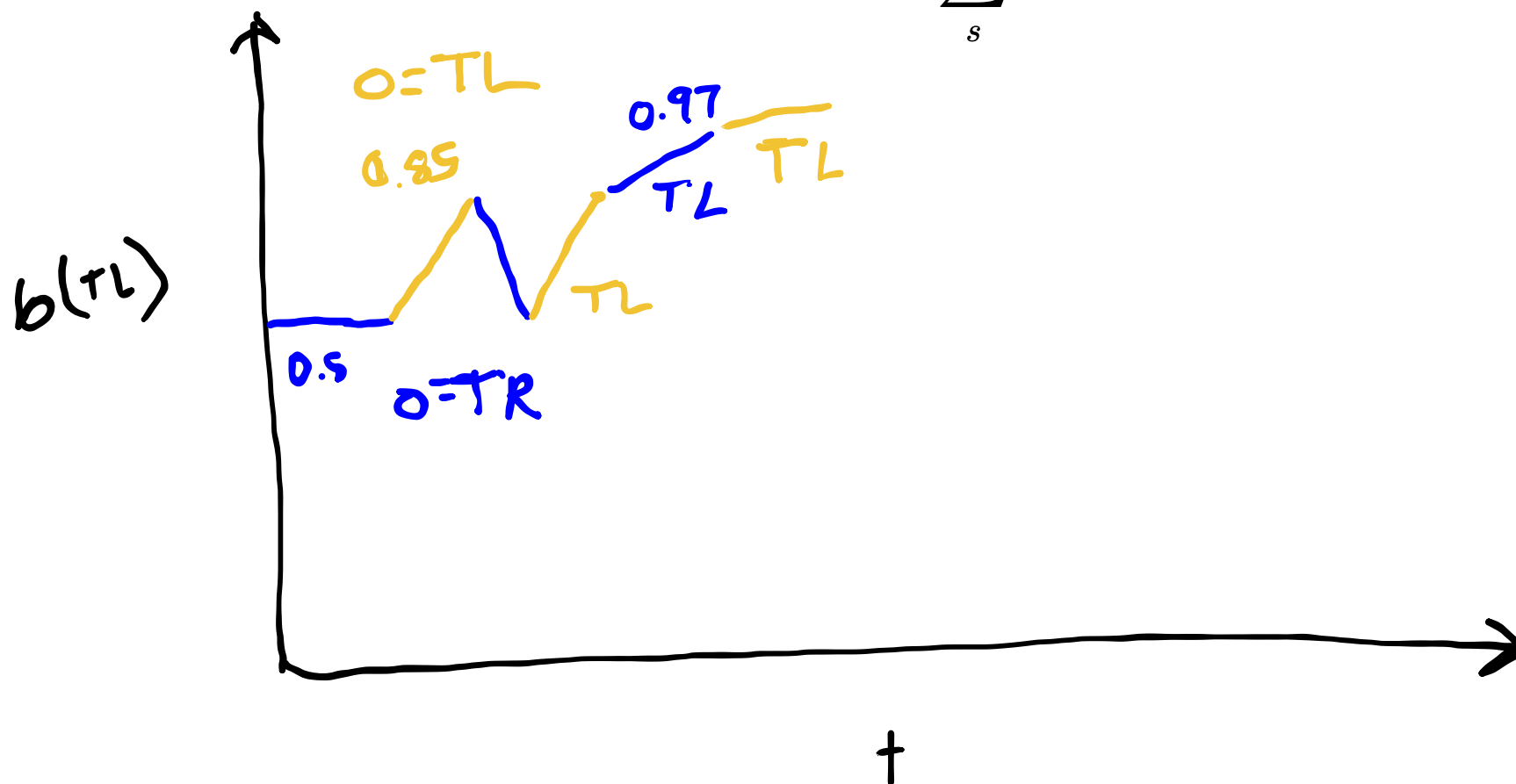
Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



Belief Dynamics

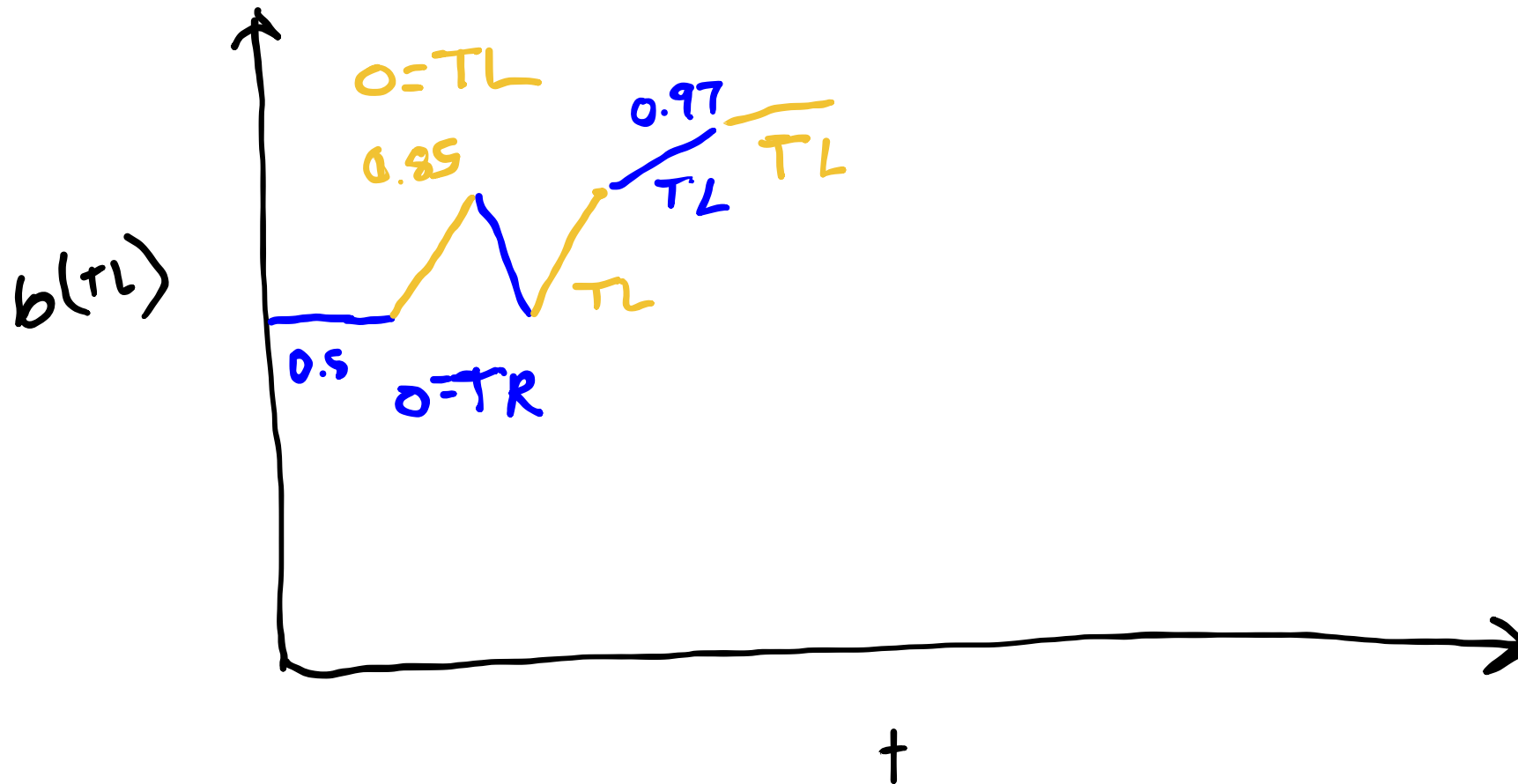
$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



One-step utility

α -vectors define a policy

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



Exercise 2: Crying Baby 1-Step Utility

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

Draw the 1-step utility α -vectors
for the Crying Baby problem.

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

Alpha Vectors for Conditional Plans

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (oL) (oR)

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

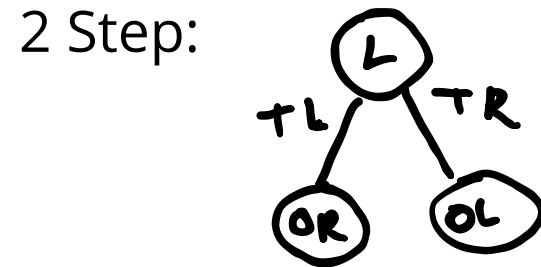
1 Step:   

2 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

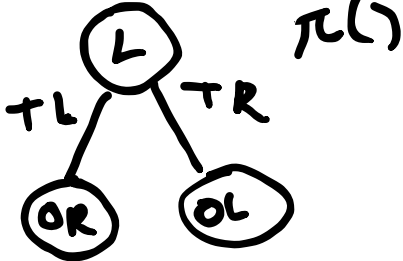
1 Step: (L) (OL) (OR)



Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

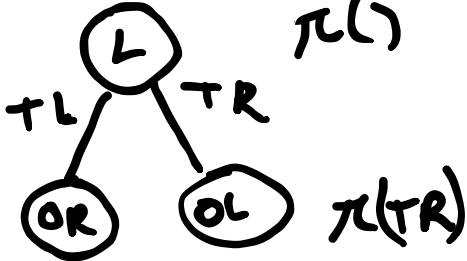
1 Step: (L) (OL) (OR)

2 Step: 

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

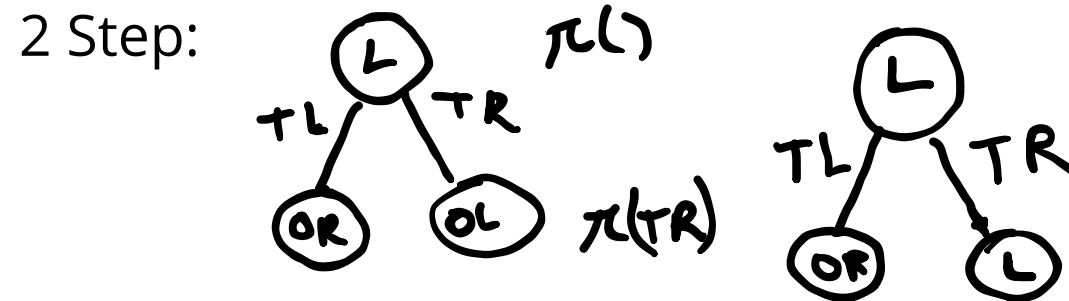
1 Step: (L) (OL) (OR)

2 Step: 

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

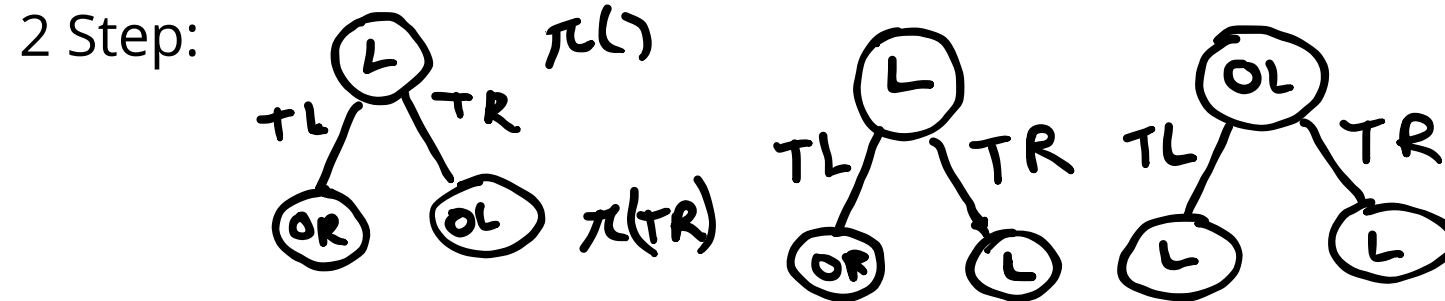
1 Step: (L) (OL) (OR)



Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)

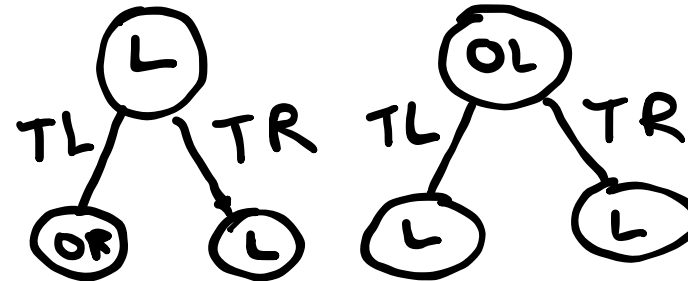


Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)

2 Step: (L) $\pi(L)$
TL TR
(OR) (OL) $\pi(TR)$

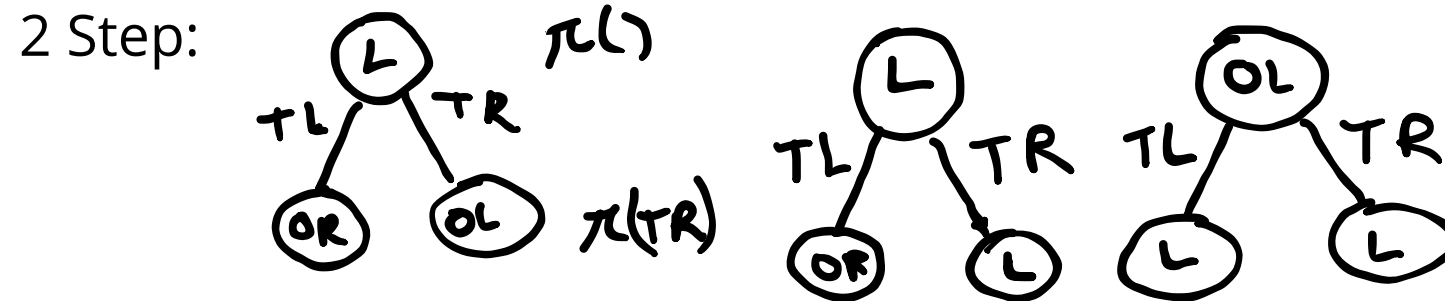


$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)



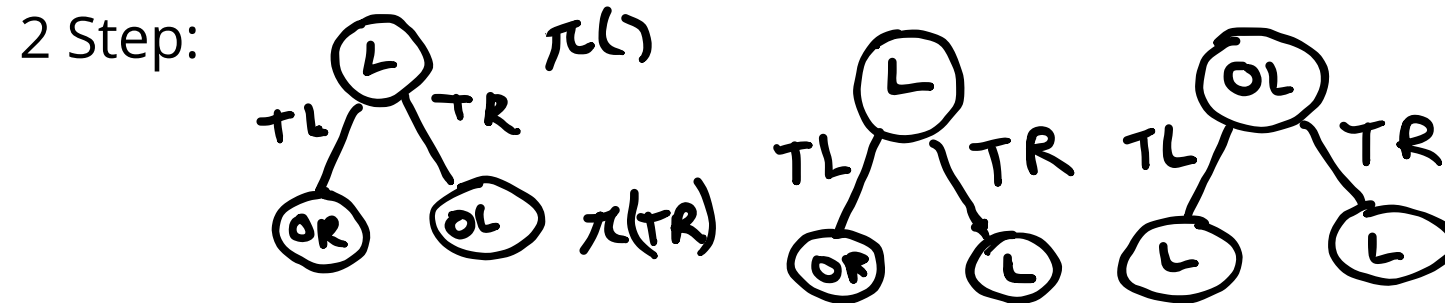
$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

27 two step plans!

Alpha Vectors for Conditional Plans

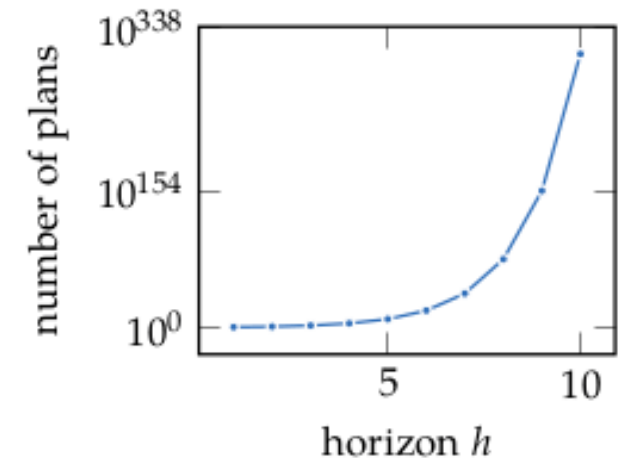
Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)



$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

27 two step plans!



Alpha Vectors for Conditional Plans

Alpha Vectors for Conditional Plans

For 1-step: $U^\pi(s) = R(s, \pi())$

Alpha Vectors for Conditional Plans

For 1-step: $U^\pi(s) = R(s, \pi())$

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o O(o | \pi(), s') U^{\pi(o)}(s') \right]$$

POMDP Value Functions

POMDP Value Functions

$$V^*(b) = \max_{\alpha \in \Gamma} \alpha^\top b$$

Exercise: 2-Step Crying Baby α Vectors

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

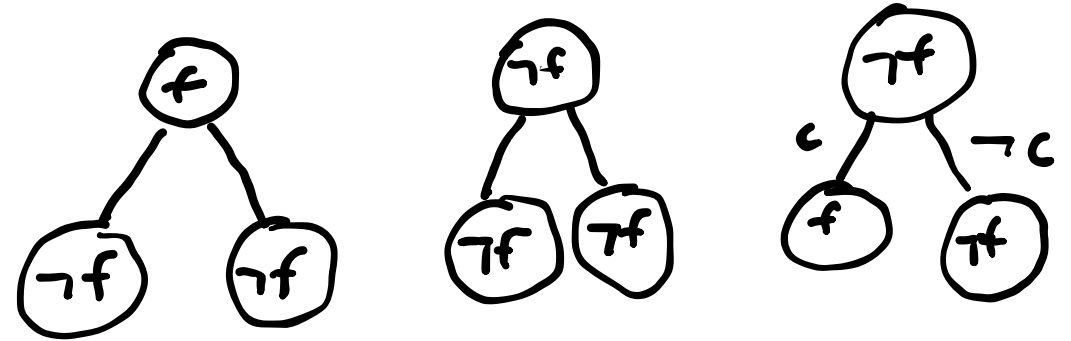
$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

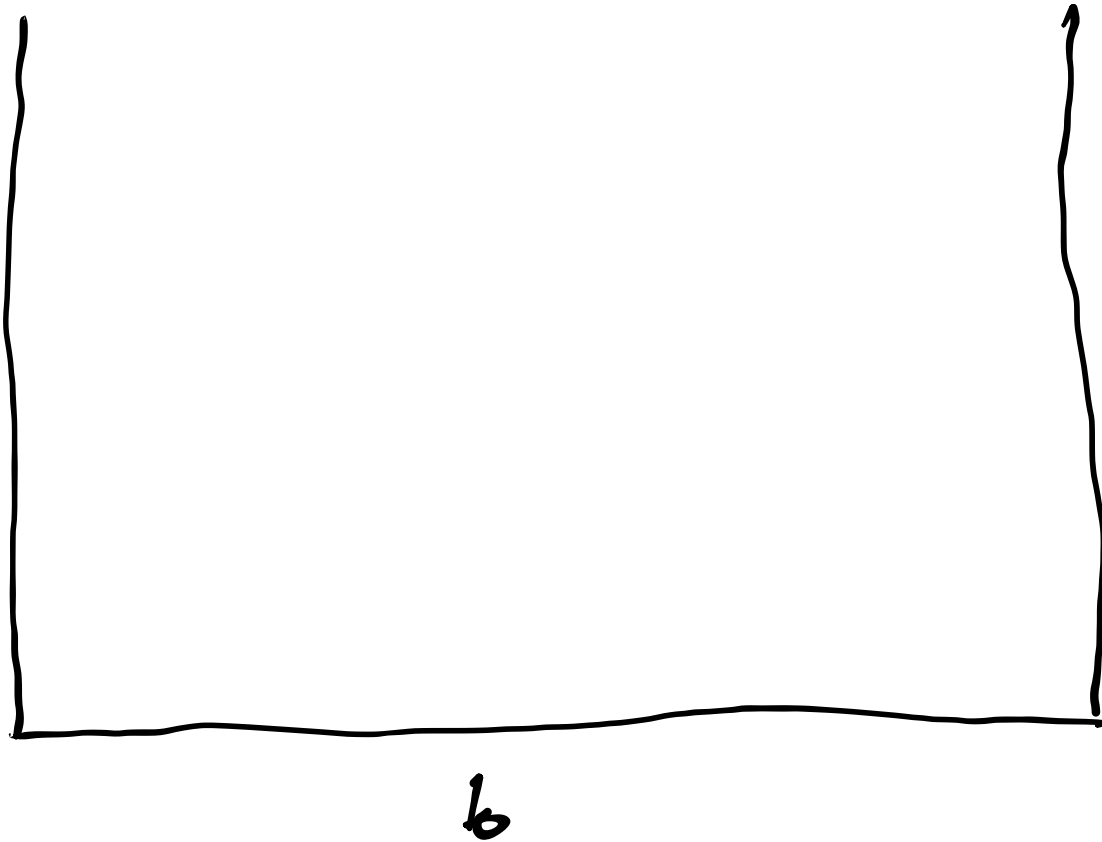
$$\gamma = 0.9$$



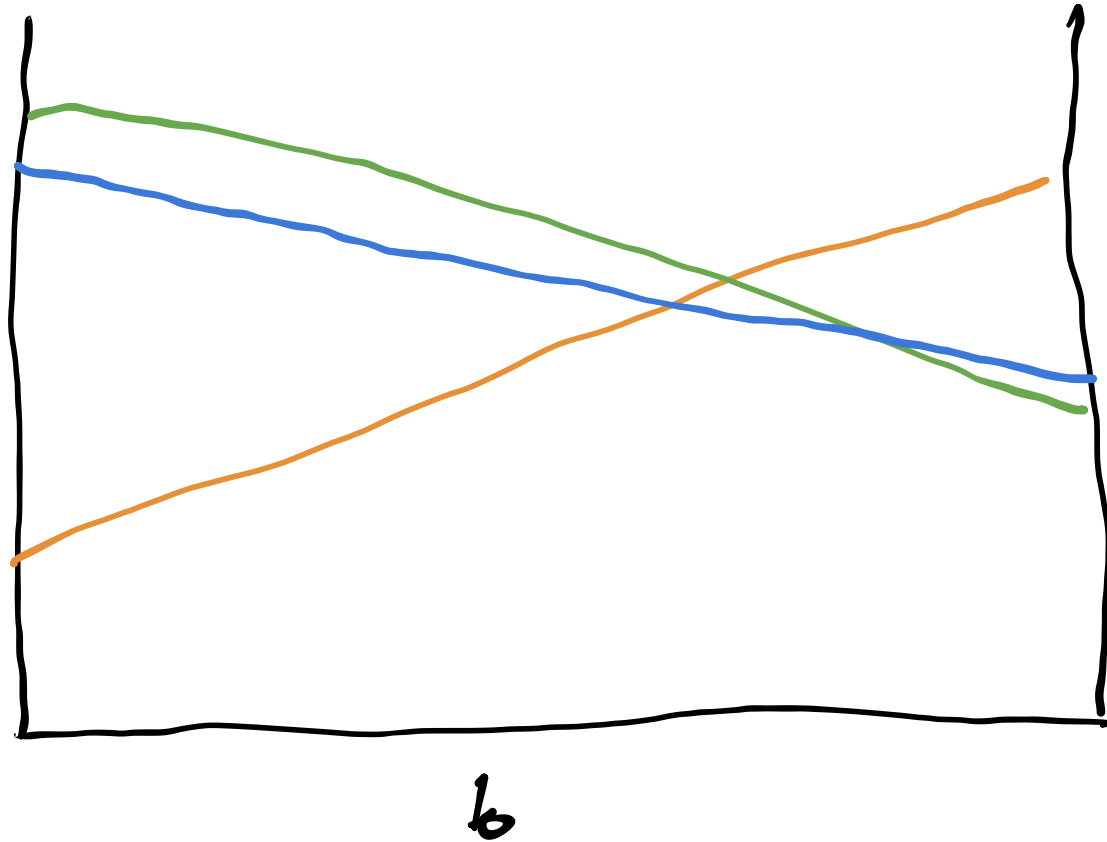
$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

α -Vector Pruning

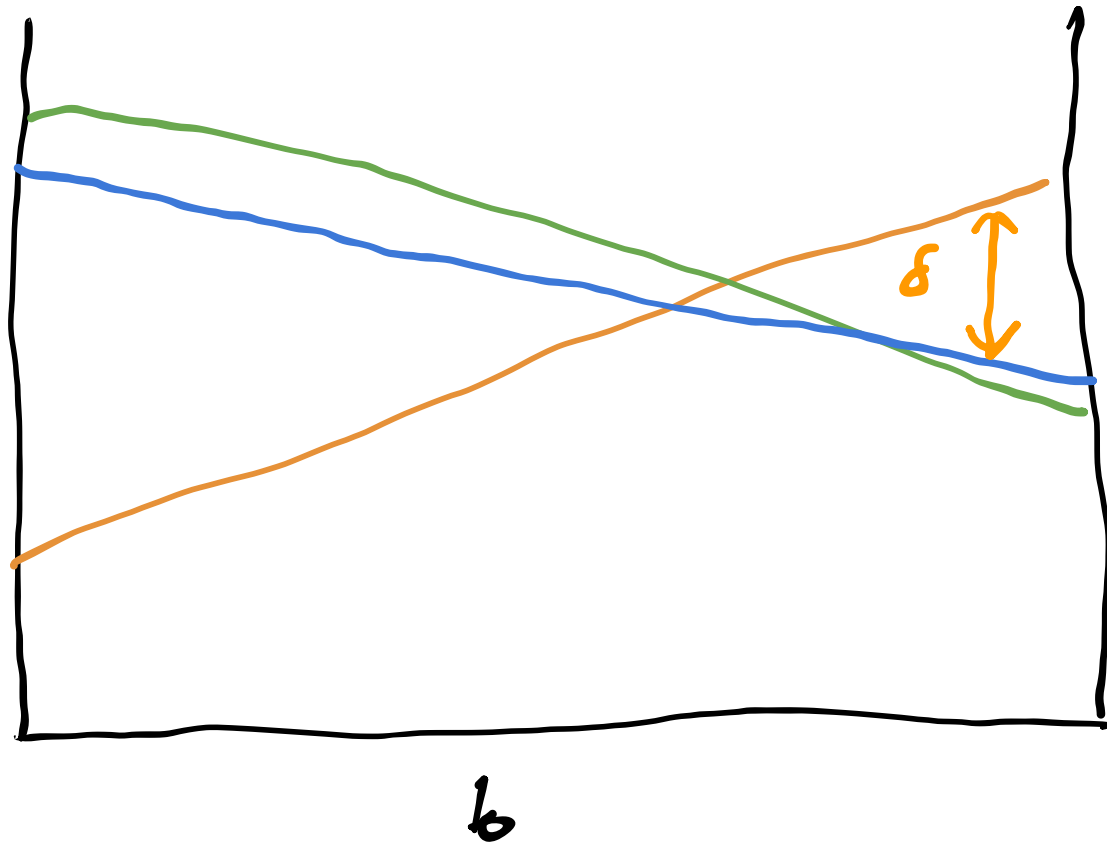
α -Vector Pruning



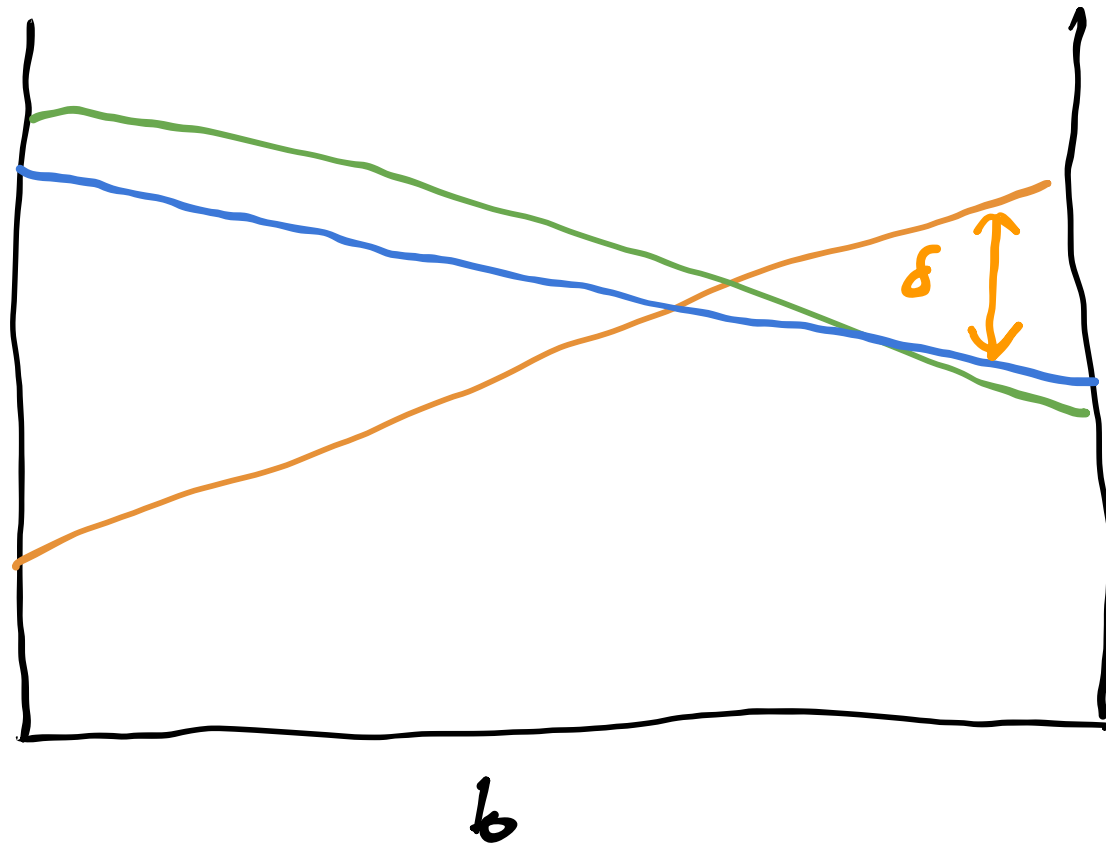
α -Vector Pruning



α -Vector Pruning

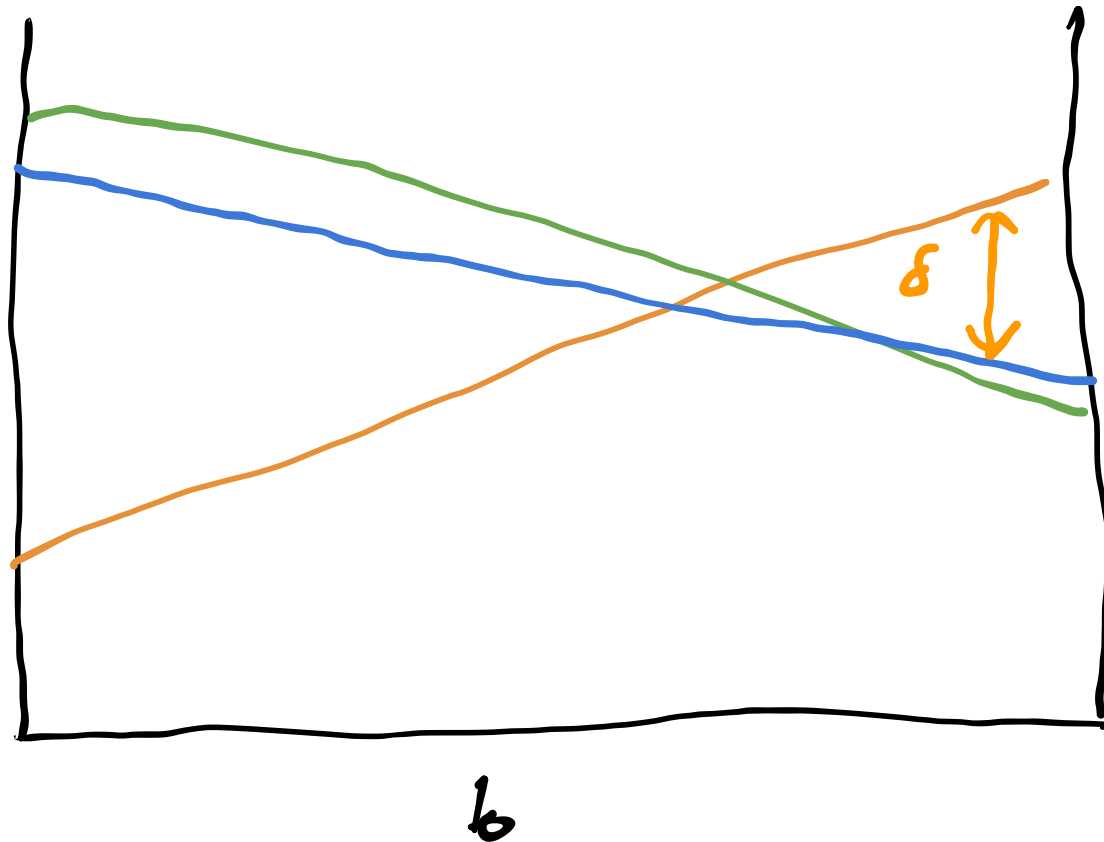


α -Vector Pruning



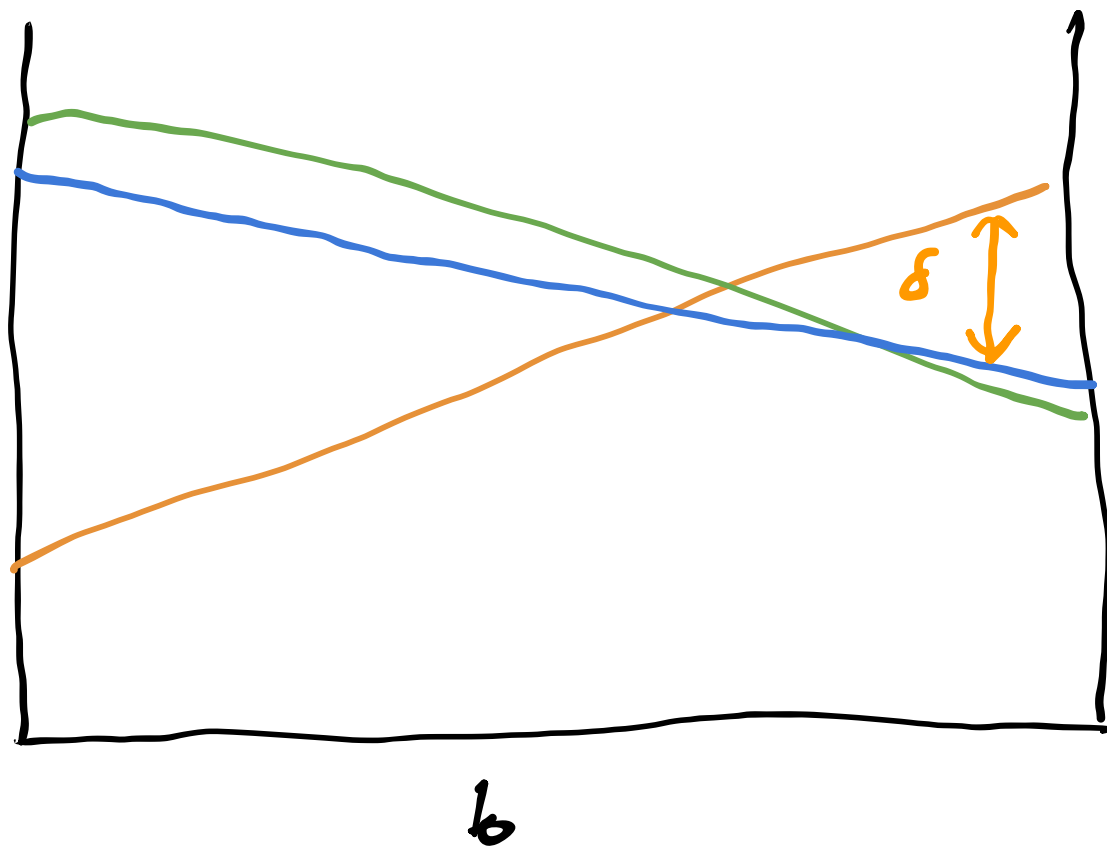
maximize δ
 δ, b

α -Vector Pruning



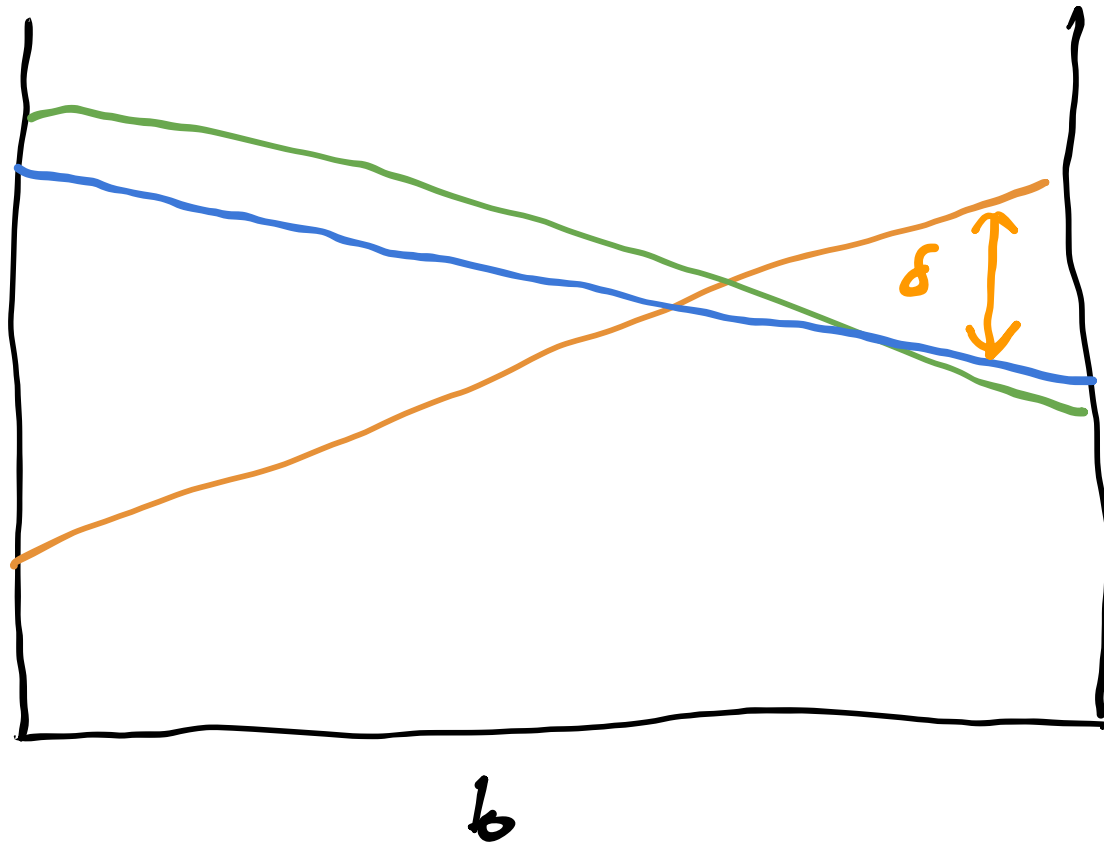
maximize δ
subject to $b \geq 0$

α -Vector Pruning



$$\begin{aligned} &\underset{\delta, b}{\text{maximize}} && \delta \\ &\text{subject to} && b \geq 0 \\ &&& \mathbf{1}^\top b = 1 \end{aligned}$$

α -Vector Pruning

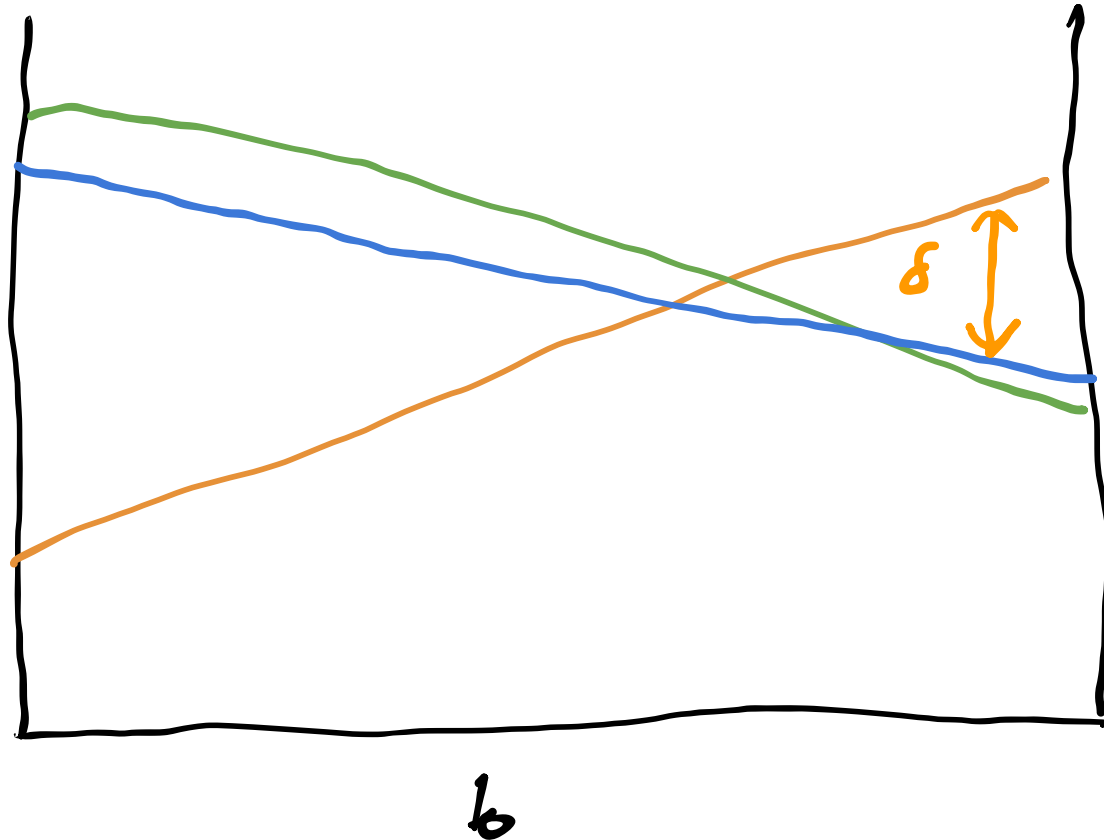


maximize δ
subject to $b \geq 0$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

α -Vector Pruning



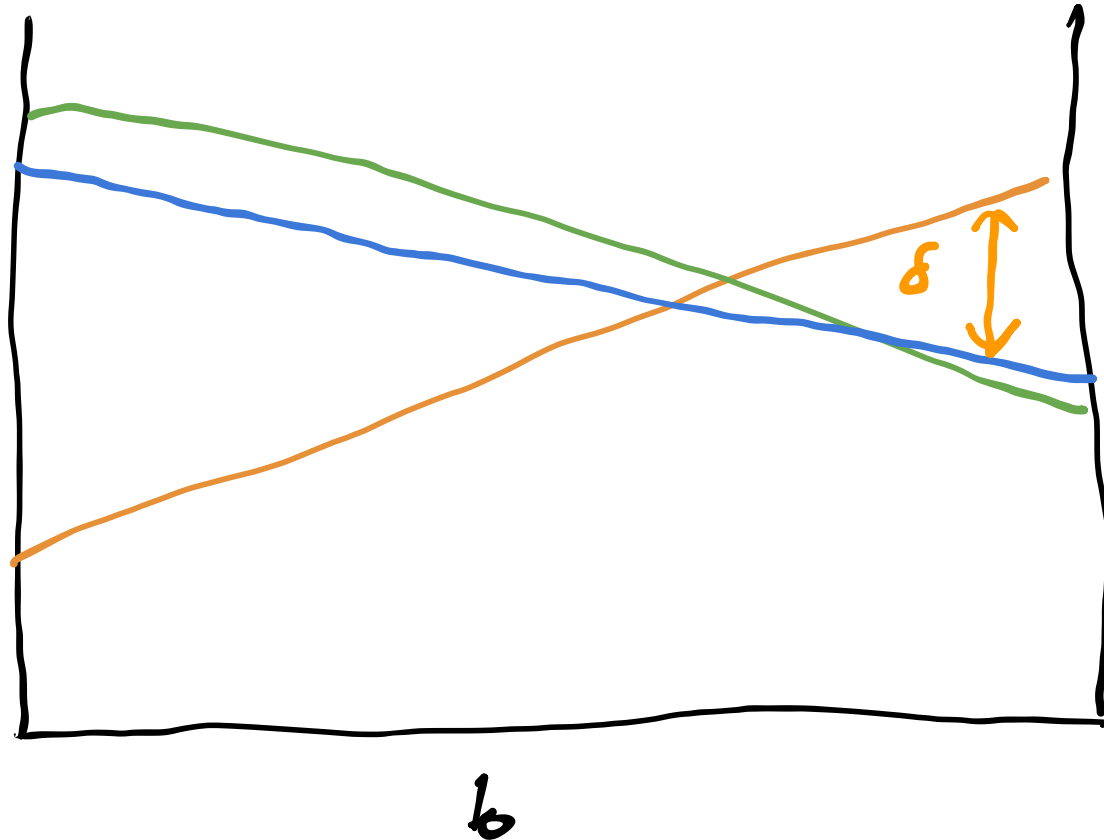
maximize δ
subject to $b \geq 0$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

"Linear Program": easy
optimization problem

α -Vector Pruning



$$\begin{array}{ll} \underset{\delta, b}{\text{maximize}} & \delta \\ \text{subject to} & b \geq 0 \end{array}$$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

"Linear Program": easy
optimization problem

If $\delta^* > 0$, α is not dominated; b
solution sometimes called
"witness".

POMDP Value Iteration (horizon d)

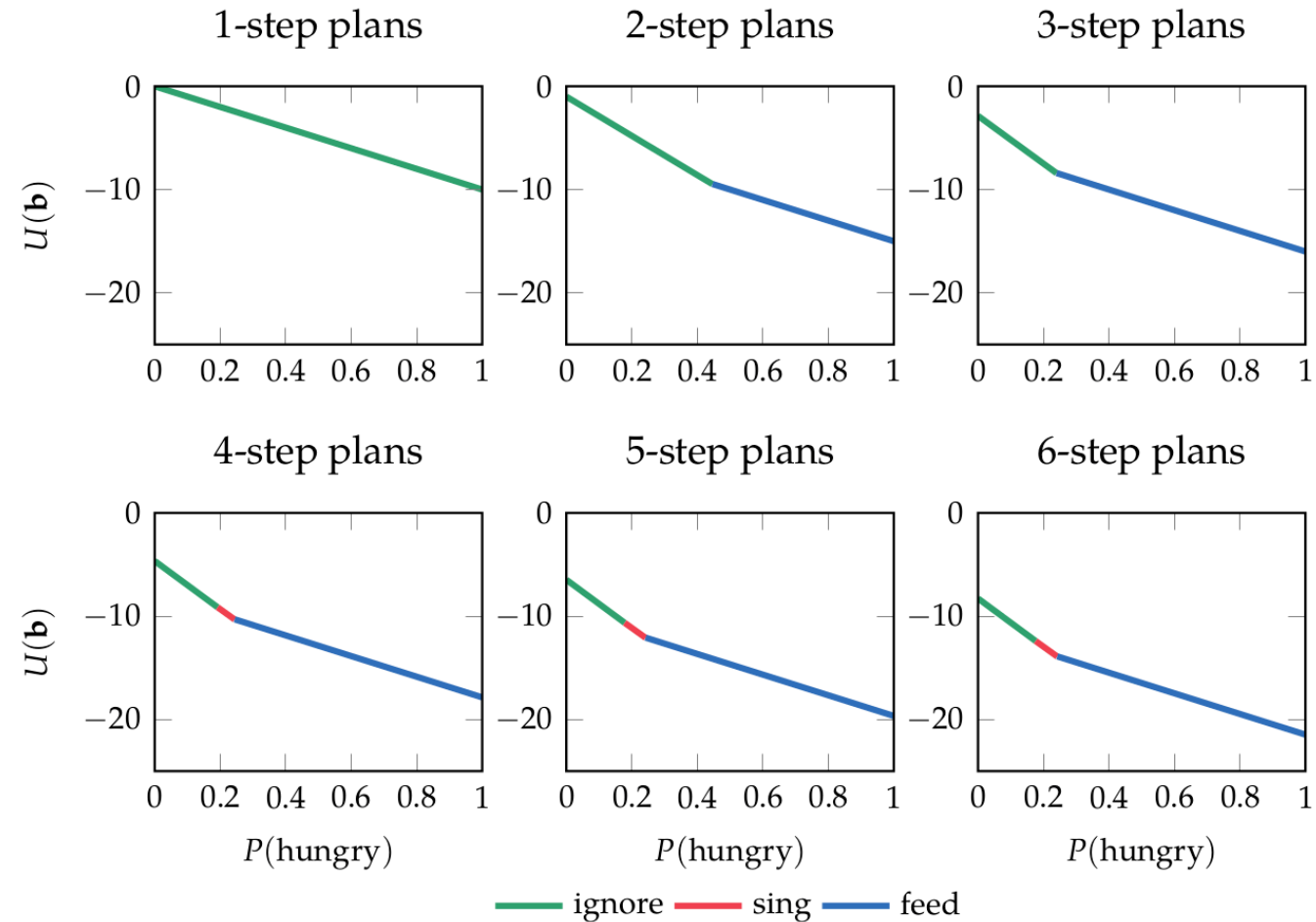
$\Gamma^0 \leftarrow \emptyset$

for $n \in 1 \dots d$

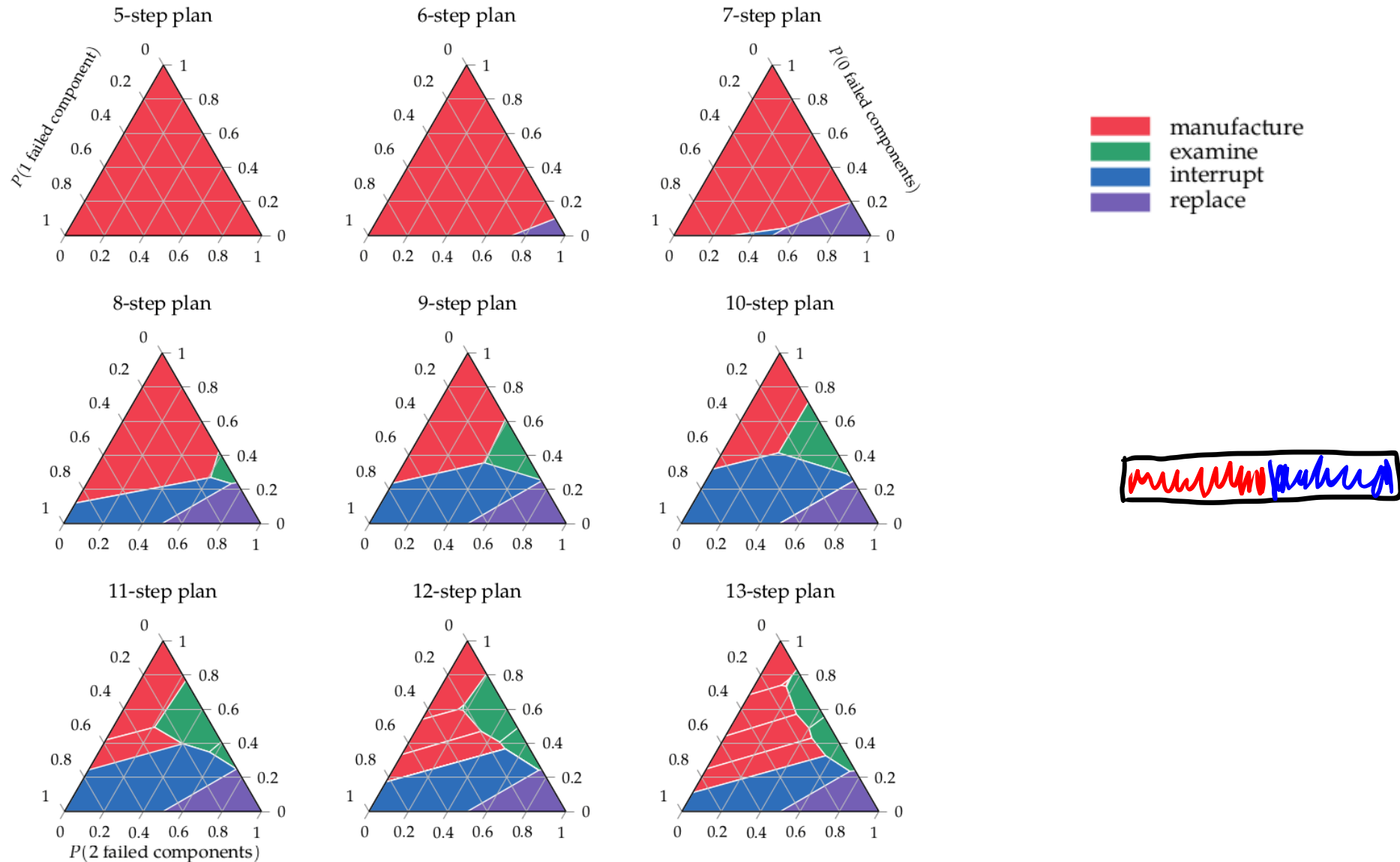
 Construct Γ^n by expanding with Γ^{n-1}

 Prune Γ^n

Finite Horizon POMDP Value Iteration



Finite Horizon POMDP Value Iteration



Recap

Recap

- A POMDP is an MDP on the _____

Recap

- A POMDP is an MDP on the belief space

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a conditional plan